Effect of Medicaid Expansion on Insurance Coverage among Foreign-born Adults

Shadi Seyedi

University of Maryland Baltimore County

Author Note

Correspondence concerning this article should be addressed to Shadi Seyedi, E-mail: sseyedi1@umbc.edu

Effect of Medicaid Expansion on Insurance Coverage among Foreign-born Adults

Introduction

The Affordable Care Act (ACA) was passed in 2010, aiming to reform healthcare and provide nearly universal health insurance coverage for low-income people. The law introduced a series of reforms, including insurance market regulations, the creation of health insurance exchanges, and income-based subsidies to address persistent gaps in insurance coverage and inefficiencies in the U.S. healthcare system (C. Courtemanche et al., 2017).

One of its most significant provisions was the expansion of Medicaid eligibility to include low-income adults without dependent children, covering individuals with incomes up to 138 percent of the federal poverty level (FPL). Although the expansion was designed to be implemented nationwide, the Supreme Court's 2012 ruling rendered it optional for states, delaying its full rollout until 2014 (Kaiser Family Foundation, 2022).

A decade later, the uninsured rate has fallen to historic lows, yet 25.3 million non-elderly Americans remained uninsured in 2023 (Tolbert et al., 2024). Immigrants bear a disproportionate share of this burden; they account for nearly one-third of all uninsured people—about 8.6 million in 2023 (Buettgens & Ramchandani, 2023). Whereas, foreign-born individuals, commonly referred to as immigrants, make up approximately 14-16% of the 2021 U.S. population, as reported by Schumacher et al. (2023) and Straut-Eppsteiner (2023). This diverse demographic includes naturalized citizens, permanent residents, visa holders, and undocumented immigrants. 55% of adult U.S. immigrants have naturalized citizenship. Among the remaining foreign-born individuals, 24% are permanent residents and 5% are other lawfully present immigrants, while the remaining 23%-26% are undocumented (Straut-Eppsteiner, 2023).

While low-income undocumented immigrants are ineligible for Medicaid, a significant portion of low-income foreign-born individuals qualify for public program

benefits. Eligibility criteria typically extend to naturalized citizens and other foreign-born non-citizens with lawful U.S. residency of at least five years, in accordance with government policies like the Personal Responsibility and Work Opportunity Reconciliation Act (PRWORA) of 1996 and the Deficit Reduction Act of 2005 (Bustamante et al., 2019).

Despite their legal status, immigrants are more likely to be uninsured due to their representation among low-income groups (Hoynes et al., 2006), and their limited access to employer-provided health insurance (Chang, 2019). For those eligible for public programs such as Medicaid, immigrants could face unique challenges in accessing them due to their language barriers and familiarity with the US healthcare system and fear of jeopardizing immigration status(Carrasquillo et al., 2000).

Within this dynamic landscape of immigration policies and demographic shifts, it is crucial to assess whether Medicaid expansion effectively addresses healthcare disparities and access challenges faced by foreign-born individuals. Without such an examination, policymakers and healthcare stakeholders lack essential insights to tailor interventions, improve healthcare equity, and ensure that Medicaid expansion benefits all vulnerable populations.

Literature review

Extensive research has been conducted by policy analysts and researchers, exploring the multifaceted effects of Medicaid expansion on various outcomes (Mazurenko et al., 2018). These investigations span a wide spectrum, from highly expected outcomes such as insurance coverage, health care access (C. Courtemanche et al., 2017, 2018; C. J. Courtemanche et al., 2023; Miller et al., 2021), health care utilization (Toyoda et al., 2020), , and variety of health outcome (Angier et al., 2020; De, 2021; Miller & Wherry, 2017; Miller et al., 2021; Soni et al., 2020) to less expected consequences, including crime reductions (Wen et al., 2017), changes in workforce participation(Gooptu et al., 2016),and income inequality (Buettgens et al., 2021; Lin et al., 2021). Many of these studies and more have examined the distribution of Medicaid expansion's benefits across diverse

beneficiary groups, like comparing outcomes for low-income rural residents with their urban counterparts (Soni et al., 2017), examining access to care and insurance coverage among low-income people of color (Buchmueller et al., 2016; Lee & Porell, 2020; Lipton et al., 2019; Ortega et al., 2018; Park et al., 2019; Yue et al., 2018)

While the ACA's effects have been extensively studied for the general U.S. population and across various demographics, comparatively little attention has been devoted to understanding the differential impact of ACA and Medicaid expansion on foreign-born. Among those that examine the effect of ACA among immigrants, some qualitative studies have played a crucial role by offering insights into the insurance access and healthcare utilization patterns of immigrants. For instanceJoseph (2016), in their study, compare the impact of citizenship status on health coverage options under the ACA in Massachusetts, analyzing its implementation alongside the state's 2006 health reform. They argue that despite the ACA's broader healthcare access, its imperfect execution led to demographic disparities in coverage, highlighting the role of public policies in creating healthcare boundaries. Similarly, Marrow and Joseph (2015) studied how the ACA affected healthcare access for unauthorized immigrants in San Francisco and Massachusetts. They found that although there were some positive symbolic and social changes recognizing legitimacy, persistent financial, language, and bureaucratic barriers show that local strategies can only do so much without broader federal immigration reform. These qualitative examinations provide contextual information that enriches our understanding of the complex interplay between immigration status, healthcare policies, and outcomes, yet they lack generalizability and cannot quantify coverage effects.

On the quantitative side, several studies have examined patterns of insurance access and healthcare utilization among immigrants before and after the ACA, often using state-level data. For example, Chu et al. (2021) investigates the effects of the ACA on healthcare access for White, Asian, and Latino immigrants in California, while Porteny et al. (2022) analyzes its impact across a range of immigrant subgroups. Similarly, Pandey

et al. (2014) evaluates the ACA's implications for immigrant access to care in New Jersey. Notably, these studies consistently report coverage gains across all groups, though immigrants tend to experience a slower pace of improvement. However, their focus on individual states limits the generalizability of their findings, making it difficult to assess whether similar patterns hold nationwide.

A smaller set of studies has moved to use nationally representative data. Most are descriptive before–after comparisons. Using 2013–2015 MEPS, Tarraf et al. (2021) document post-ACA gains in coverage for U.S-born and foreign-born adults but, because they do not distinguish expansion from non-expansion states, cannot attribute changes to Medicaid policy. Bustamante et al. (2019) conduct a similar before-and-after analysis with NHIS, comparing immigrant and native coverage trends from 2011 to 2017 Likewise, Li et al. (2021) employ MEPS to compare pre- and post-ACA health care access, but do not exploit policy variation across states. Although these studies confirm substantial overall gains, they offer limited causal inference and provide little insight into whether Medicaid expansion specifically narrowed immigrant–native gaps.

More recent quasi-experimental work has begun to exploit policy variation but still leaves important questions unanswered. Altman et al. (2023) impute legal status in ACS data and show that Medicaid expansion increased coverage for naturalized citizens and lawful permanent residents (LPRs) but not for likely unauthorized immigrants; yet their findings hinge on complex imputations that may misclassify non-LPRs. Moreover, their design does not compare immigrants with natives

The only study to examine immigrant-native differentials causally is Stimpson and Wilson (2018), who applied a triple-differences model to ACS 2010–15 and attributed roughly five percentage points of the decline in uninsurance among non-citizens to Medicaid expansion. Several limitations, however, affect their conclusion. First, the analytic window ends in 2015 —just two years after expansion—omitting later adopters and longer-run effects. Second, all noncitizens are pooled, even those in the United States

fewer than five years who remain ineligible for Medicaid, potentially diluting treatment effects. Third, the outcome is binary uninsurance; the study does not identify whether gains reflect new Medicaid take-up or shifts from other coverage sources. Fourth, their pre/post specification offers a single average treatment effect, with no examination of dynamics or sensitivity to state-level policy environments. Finally, key contextual factors—state unemployment, immigration policy climate, and political ideology—are unmeasured, raising concerns about omitted-variable bias.

The present study addresses these gaps by (i) extending the observation period through 2019 to capture both early and late adopters, (ii) restricting the sample to adult immigrants with more than 5 years of residency and adults with less than 138 percent FPL—those most clearly targeted by the policy, (iii) distinguishing Medicaid enrollment, direct purchase, employer coverage, and uninsurance to track the source of gains, (iv) estimating event-year difference-in-differences models that reveal the timing of effects, and (v) controlling for time-varying state factors such as unemployment, structural xenophobia, and political ideology. Together, these refinements yield a more precise and policy-relevant estimate of how the ACA's Medicaid expansion shaped insurance coverage for low-income foreign-born adults nationwide.

This paper asks four questions:

1. Does Medicaid expansion reduce the uninsured rate among qualified immigrants, thereby improving access to health insurance for foreign-born individuals?

2. Does Medicaid expansion increase Medicaid enrollment among eligible foreign-born individuals?

3. Are there disparities in the uninsured rate and Medicaid enrollment between foreign-born and U.S.-born individuals following Medicaid expansion?

4. Can immigrant status explain the observed disparities in insurance coverage gains among different racial and ethnic groups following the implementation of Medicaid

expansion?

In the next section, Theoretical Frameworks, I present relevant theories and explore literature to build a framework for understanding Medicaid's impact on immigrants, especially how it differs for foreign-born and native-born individuals. I discuss research design and empirical methods for robust estimates in section 3. Section 4 will present descriptive tables and regression results, followed by a discussion section and concluding remarks on policy implications, limitations, and future research suggestions.

Theoretical framework

My theoretical framework for this study is based on economics, healthcare policy, immigration studies, and health inequalities. I first apply economic theories to understand health insurance choices, especially regarding subsidies. Then, I explore theories and models of social benefit non-utilization, constructing a framework for understanding how individuals access and utilize these benefits. Within this framework, I examine the barriers and facilitators that influence the enrollment of eligible foreign-born individuals in programs like Medicaid, disentangling cause that may give rise to disparities in enrollment among diverse immigrant groups. Finally, this theoretical underpinning informs our research hypotheses, which will be evaluate in later sections.

Rationale for Enrolling in Medicaid

From a standard economic perspective, demand for health insurance hinges chiefly on price. Both conventional risk-aversion models (Pauly, 1968) and Nyman's income-transfer theory (Nyman, 2003) predict that when the out-of-pocket premium falls to zero—as it does under Medicaid for eligible low-income adults—the expected utility of coverage rises sharply, leading to near-universal take-up in the absence of other frictions.

From a standard economic perspective, the demand for health insurance is primarily price-driven. Both conventional risk-aversion models (Pauly, 1968) and Nyman's income-transfer theory (Nyman, 2003) posit that when the out-of-pocket premium cost

drops to zero—as with Medicaid for eligible low-income populations—the expected utility of coverage surges, theoretically leading to near-universal take-up in the absence of frictions.

Yet empirical evidence reveals a stark divergence from this prediction: imperfect uptake of social benefits persists even when financial barriers are eliminated (Wehby & Lyu, 2018). This gap between theory and practice highlights the role of non-price factors that disproportionately deter marginalized groups from enrolling and exacerbating social inequality among citizens (Ko & Moffitt, 2022).

Why Eligible Adults Remain Uninsured

To gain a deeper understanding of non-utilization complexities within social welfare programs like Medicaid, and understand the non-take-up paradox I delve into the various models and frameworks that have been developed to understand why eligible individuals sometimes do not enroll in such programs.

Economic Models

Economic theories provide valuable insights into the benefits of enrollment decisions. focusing on the cost-benefit trade-off. Within this framework, a rational decision-maker examines the various costs associated with program participation and weighs them against the perceived benefits to decide on enrolling in a program.To explain the lower than expected participation rate in social benefit programs, economists argue that costs associated with program participation, such as information collection, application procedures, and stigma, are substantial. Simultaneously, the expected benefits of the program are comparatively modest(Ko & Moffitt, 2022).

Psychological and Behavioral Models

A different set of models discuss the psychological and behavioral aspects of benefit take-up decisions, delving into individuals' cognitive and decision-making processes when accessing social benefits. Kerr's sequential model is one of them. Kerr (1982) identified six conditions that must be met for a take-up decision: perceived need, program understanding, perceived eligibility, benefit value, feelings about the process, and perceived

stability. These conditions delve into the psychological aspects of individuals' perceptions, beliefs, and attitudes toward benefit programs, outlining the sequential steps that individuals typically go through when deciding to access social benefits.

Integrated Model: Van Oorschot's Multilevel System

Van Oorschot's Multilevel System Model provides a comprehensive perspective on non-take-up decisions, integrating both economic and psychological elements. According to this framework, individuals need to go through three stages to take up a social benefit. Firstly, they must become aware of the program (threshold stage), then assess its value (trade-off stage), and finally, apply for it (application stage). This model recognizes the interplay of policy, administrative, and individual factors in non-take-up decisions throughout these three stages. This model distinguishes between two types of non-take-up: one where an individual does not claim a benefit as a result of an individual factor, and the other where a person claims a benefit but does not receive it due to some administrative factor (van Oorschot, 1998). Additionally, Janssens and Van Mechelen (2022) introduces another category for non-take-up in the model, which occurs when eligibility rules prevent individual access; this type of non-take-up can be attributed to the policy factors.

This framework serves as a valuable tool for understanding non-take-up in Medicaid and distinguishing unique factors that contribute to the non-take-up of native-born and foreign-born individuals.

Foreign-born Enrollment Factors and Formulating Hypotheses

The empirical evidence on social benefit uptake rates among foreign-born individuals, compared to their native-born counterparts, presents a contradictory picture. Studies vary in their conclusions, with some reporting higher, equal, or lower participation rates among immigrants. For instance, Borjas and Trejo (1991) found a higher involvement of immigrants in social welfare in 1980 than natives, attributing it to the Magnet hypothesis suggesting that the state's generous welfare offer attracts low-skilled immigrants. However, subsequent studies, including (Giulietti, 2014), question the validity

of this hypothesis. Similarly, Bird et al. (1999) indicate, in their study, that immigrants are not more inclined than natives to seek welfare assistance; their likelihood is equal. The argument suggests that the perception of a higher immigrant participation rate might stem from their increased likelihood to live in situations that qualify them for welfare. On the other hand, a more recent wave of research indicates a lower participation rate among immigrants in social programs like SNAP and Medicaid (Ku & Bruen, 2013; Skinner, 2012).

I utilize the modified Van Oorschot model as a guiding framework in my literature exploration of factors contributing to non-take-up. The objective is to comprehend the disparities in social benefit take-up between the foreign-born and native populations, enabling the effective formulation of my research hypotheses and the creation of the underlying path diagram.

I therefore categorize the identified variables into three levels: client level, administration level, and system level. This classification enables me to differentiate between different sorts of non-take-up and facilitates my understanding of the existing situation

Client Level

Several studies have emphasized the significance of English proficiency, which significantly shapes immigrants' perspectives on the welfare state and explains the pattern of health care access and health status. (Adigun et al., 2021; Clarke & Isphording, 2017; Ryan et al., 2008; Tumlin & Zimmerman, 2003). Proficiency in English enhances immigrants access to a greater variety of information and opportunities; it helps them grasp the existence of public benefits and their eligibility restrictions, generally reducing the cost of acquiring information as well as the process cost, such as difficulties in completing paperwork or interacting with administrative staff. Individuals who do not feel comfortable speaking and writing in the language of the application process may require additional assistance, thereby incurring higher process costs.

Another factor that could differently affect the uptake of benefits among foreigners

is their social/ethnic network. Numerous studies have emphasized the significant influence of immigrants' social networks on their engagement with social welfare programs (Furtado & Theodoropoulos, 2016; Gee & Giuntella, 2011). Social networks can impact participation through the transmission of social norms and information exchange. For example, individuals with limited language skills may offset higher information and processing costs by seeking assistance from their social networks. Another way in which networks could affect uptake is through the dynamics of stigma and assimilation. Remaining within one's ethnic group may reduce the likelihood of assimilation into the host culture, potentially resulting in foreign-born individuals facing distinct stigma costs compared to natives.On the other hand, the duration of residence in the host country can impact access to public benefits (Choi et al., 2016). The longer foreign-born individuals live in the U.S. (more than 10 years), the more likely they are to know about and use public benefits. This is because, over time, they become more aware of the available support.

Administration level

When considering the factors influencing administrators' decisions to erroneously reject eligible individuals, research find that administrators have a significant degree of discretion, which can impact service delivery (Kline, 2019). This aligns with Lipsky's theory of street-level bureaucracy, which argues that public service employees responsible for benefit distribution have substantial discretionary power (Lipsky, 2010). Due to time and information constraints, administrators may develop routines and streamline processes that can affect the outcomes of their services. Brodkin and Majmundar (2010) extended this theory to examine the impact of administration on the non-take-up of TANF (Temporary Assistance for Needy Families). They found that complex procedures and excessive bureaucracy could result in non-acceptance due to the higher risk of errors. Administrators may create routines and processes that impose varying hidden costs on clients' decision-making processes. Given the complexity of Medicaid eligibility requirements for immigrants under the Medicaid expansion scenario, these administrative

burdens may be more pronounced.

To shed light on administrator regulation regarding the non-acceptance of immigrants, Mallet and Garcia (2021) conducted a comparative and ethnographic investigation. Their interviews with 82 eligible adults who were denied welfare assistance by administrators revealed instances of administrator prejudice based on classicism toward immigrants and ethno-racial disparities. This prejudice could lead to low participation rates among immigrants, as administrators have extensive authority and can reject applications based on their perception of ineligibility, even when applicants are eligible. In many cases, immigrants were required to submit additional documentation and encountered discrimination from administrators. While social workers play a significant role in the lives of immigrants and their influence on welfare participation should not be underestimated, accounting for these factors can be challenging in research.

System level

The design of social policies can have a substantial impact on the information, processing, and behavioral costs borne by clients and on the administrative burden, potentially increasing the likelihood of errors. Selectivity in policy design, which targets specific population subgroups, can significantly influence adoption behavior. For instance, in the case of Medicaid expansion, undocumented immigrants and those with less than five years of residency are typically ineligible for program benefits. Highly selective social programs are prone to fragmentation, leading to information overload and increased information costs for clients. Targeting may also heighten the perception of stigma among certain groups (Janssens & Van Mechelen, 2022; Lucas et al., 2021).

Another element to consider is the chilling effect, which can lead to reduced program participation among eligible groups due to confusion, concern, and fear of violating laws or regulations (Gutierrez, 2021). This effect is particularly relevant in the context of Medicaid enrollment and can be influenced by the following:

Public Charge Rule: The Department of Homeland Security's Notice of Proposed

Rulemaking under section 212(a)(4) of the Immigration and Nationality Act of 2018, also known as the Public Charge rule, had the potential to deter immigrants. It stipulated that legally-presented immigrants seeking permanent residency could be labeled as public charges and denied permanent residency if they used public benefits, including Medicaid (Khullar & Chokshi, 2019).

Immigration Enforcement Measures: Measures such as workplace raids, deportations, and restrictive state/local immigrant integration laws can increase fear and anxiety among legal and undocumented immigrants. This anxiety can lead individuals to avoid activities involving public officials, including seeking healthcare services. Stress and anxiety related to undocumented or unauthorized status and the fear of deportation are not limited to undocumented immigrants alone; they also affect lawful permanent residents. The stress and anxiety associated with race/ethnicity-related concerns about lawful presence in the United States are widespread (Perreira & Pedroza, 2019).

Building upon my understanding of barriers and facilitators within Van Oorschot's framework, I formulate the following research hypotheses:

1. I hypothesize that the implementation of Medicaid expansion will result in a significant reduction in uninsured rates among both foreign-born and U.S.-born individuals.

2. I hypothesize that the implementation of Medicaid expansion policies will lead to a significant increase in Medicaid coverage rates among both foreign-born and U.S.-born individuals.

3. I expect that the impact of Medicaid expansion on the uninsured status and Medicaid coverage of eligible individuals will exhibit variation between foreign-born and U.S.-born populations, with foreign-born being less likely to benefit from expansion. These differences in impact may be attributed to unique healthcare access barriers, socio-economic factors, and other population-specific factors. I anticipate

that controlling for these factors in the analysis will mitigate the differences in the effect of Medicaid expansion between foreign-born and U.S.-born individuals.

Data

This study relies on two primary data sources to assess the impact of Medicaid expansion on immigrant insurance coverage in the United States. The American Community Survey (ACS), accessible via Public Use Microdata Sample (PUMS), is our primary data source. The ACS, conducted annually, provides a wealth of data on the U.S. population, with a substantial sample size (around 3 million individuals annually), high response rates (92% to 98%), and national and state-level coverage. It includes detailed information on health insurance, income, citizenship status, country of origin, and socio-economic demographics (Dillingham et al., 2020; US Census Bureau, 2023). However, the downside of this data set is that it does not collect data on immigrants' documentation status, preventing differentiation between the effects of Medicaid expansion on undocumented and documented immigrants.

In addition, the Kaiser Family Foundation's (KFF) State Health Facts project is utilized to determine which states adopted Medicaid expansion and when. State Health Facts, a KFF initiative, offers comprehensive, up-to-date information on over 800 health indicators for all 50 states and the District of Columbia (State's Health Fact, 2013).

Analytical Sample

To identify my study population, I carefully considered various factors, including prior research findings and Medicaid eligibility requirements. My sample includes individuals aged between 26 and 65 who earn income at or below 138% of the federal poverty level, aligning with Medicaid's income eligibility threshold under the ACA. While the age eligibility for Medicaid typically starts at 21, to avoid the potential impact of another provision of the ACA, which allowed young adults to remain on their parents' Medicaid plan until age 26, following Lin et al. (2021) I set the lower age limit at 26. These

criteria have been chosen to ensure that our analysis focuses on individuals who are most likely to qualify for Medicaid coverage.

Moreover, this study covers the period from 2011 to 2019, minimizing potential impacts from ACA provisions introduced in 2010 (C. Courtemanche et al., 2017) and complexities introduced by the 2020 COVID-19 pandemic. Addiontaly, I excluded individuals residing in specific settings[1], those with limited poverty data, and residents of Oregon, Massachusetts, Delaware, New York, and Vermont [2] due to prior policy implementations that could confound the results (Zerhouni et al., 2019).

Selecting a representative sample of non-citizens presented challenges due to varying Medicaid eligibility rules tied to immigration status. For instance, undocumented immigrants are ineligible for Medicaid, and most low-income legal residents face a five-year waiting period before qualifying. Refugees and asylees, however, are exempt from this waiting period and typically qualify immediately upon arrival (Adunlin et al., 2019).

Since our dataset lacked details on immigration status (e.g., visa type, documentation status), I addressed this by excluding non-citizens residing in the U.S. for less than five years. This exclusion aimed to approximate federal eligibility criteria, creating a more homogeneous sample of individuals likely to meet Medicaid's five-year threshold for legal permanent residents. However, this method has important limitations: It excludes refugees and asylees (who qualify immediately, regardless of time in the U.S.) and may still include long-term non-citizens who remain ineligible due to their undocumented status.

---

[1] Individuals residing in prisons, nursing homes, mental hospitals, military barracks, group homes, missions, or shelters

[2] In 2006, Massachusetts enacted a health reform bill to boost insurance coverage, setting a precedent for the ACA Medicaid expansion. Oregon expanded Medicaid in 2008 through a lottery among eligible citizens. Delaware, Massachusetts, New York, and Vermont offered Medicaid or comparable coverage to adults with incomes up to 100% of the federal poverty line or higher from 2010 to 2013.

Measures

Outcome and treatment

The outcome variables of interest include Medicaid coverage and uninsured status. Medicaid insurance coverage is captured through a binary variable, indicating whether the individual possessed Medicaid at the time of the interview. Similarly, uninsured status is a binary variable derived from ACS responses on being insured at the time of the interview. The main treatment variable examined is Medicaid expansion, which is a binary variable assigned a value of 0 for states that did not enact Medicaid expansion and 1 for states that decided to expand their Medicaid between 2014 and 2020[3]. To enhance the robustness of the analysis, I further assess the impact of Medicaid expansion on two binary insurance coverage outcomes: directly purchased insurance and employer-sponsored health insurance. Both variables were measured based on whether individuals reported holding the respective coverage type at the time of the survey interview. This approach isolates distinct pathways through which Medicaid expansion might affect coverage—either by substituting for private insurance or complementing employer-based options—while mitigating potential measurement error associated with broader coverage categories.

Covariates

I incorporate numerous covariates, selecting them based on both their theoretical relevance and predictive power. The discussion of covariates is limited to those included in the analysis models predicting Medicaid coverage and the uninsured rate. These covariates include race and ethnicity (White, Black, Asian, Hispanic, or other race), age, sex, marital status (married or single), education level (less than high school, High school, College degree, Some college or associate degree, or graduate and beyond), employment status (employed, unemployed, or not in the labor force), English proficiency, disability status (yes or no), poverty status (income 100 to 138 poverty or income below 100 poverty),

---

[3] For further information check KFF Status of state Medicaid expansion decisions: Interactive map.

percentage of life lived in the US, non-citizen status (yes or no).

I also incorporated various state-level time-varying geographic environmental variables as control factors, including the Bureau of Labor Statistics' annual state unemployment rate. Structural xenophobia, defined as discriminatory practices against immigrants in interconnected systems, was measured using the Immigration Policy Climate (IPC) index developed by Samari et al. (2021). This index assesses exclusionary and inclusionary policies across domains such as higher education, labor, driver's licenses, and immigration enforcement. State-level political ideology, evaluated through Americans for Democratic Action (ADA) scores, followed the method used by Lee and Porell (2020). ADA scores gauge how each U.S. senator from each state voted on various legislative issues, and the average score for each state serves as a proxy for its political leanings

Empirical Strategies

Static Two-Way Fixed Effects Model

To identify the impact of Medicaid expansion on insurance coverage and its differential effect among the foreign-born and US-born, I employ a difference-in-difference strategy. Because states adopted expansion in different years after 2014, I first estimate the effect of Medicaid expansion on desired outcomes with the static two-way fixed effect model (TWFE), followed by a dynamic DiD event-study analysis. Both methods account for the difference in the timing of Medicaid expansion.

Intuitively, this static TWFE model is a generalization of the 2x2 DiD design, extending beyond the two time periods of "before treatment" and "after treatment" to accommodate multiple time periods. The following econometric specification presents the main identification for this strategy.

$$Y_{ist} = \alpha + \gamma \cdot T_{st} + \beta_1(FB_{ist} \cdot T_{st}) + FB_{ist} + \beta X_{ist} + \theta(FB_{ist} \cdot C_{ist}) + \delta_s + \lambda_t + \epsilon_{ist} \quad (1)$$

In this equation, Y_{ist} represents the outcome of interest for an individual i in year t

in state s, The main parameter of interest, γ, represents the average treatment effect of Medicaid expansion. The treatment variable T_{st} is a binary indicator equal to 1 if Medicaid expansion was in effect in the state s during the year t and 0 otherwise. The key interaction term $\beta_1(FB_{ist} \cdot T_{st})$ estimates the differential effect of expansion for foreign-born individuals. The vector series (X_{ist}) includes individual-level covariates, such as age, sex, marital status, education, poverty status, English proficiency, disability status, and insurance-relevant immigration characteristics (foreign-born status, non-citizenship, and percentage of life lived in the U.S.). Additionally, I control for time-varying state-level factors—state unemployment rate, political ideology (ADA score), and structural xenophobia (IPC index)—which are commonly used to account for contextual influences on healthcare access and utilization. To capture heterogeneity in covariate effects by nativity, the model includes interaction terms $\theta(FB_{ist} \cdot C_{ist})$ where vectors C_{ist} include a subset of covariates hypothesized to operate differently for foreign-born individuals (e.g., education, marital status, race). These interactions allow for a more flexible specification that recognizes structural and contextual differences in how Medicaid expansion may affect immigrants. δ_s captures state-fixed effects and λ_t accounts for year-fixed effects.

For each year t, the control group consists of (1) low-income adults in the 14 states that never implemented Medicaid expansion during our study period (2011-2019) and (2) low-income adults in expansion states during their pre-expansion years only. Once a state expands Medicaid, all subsequent observations from that state are coded as treated $(T_{st} = 1)$ and are no longer included in the control group.

The model's standard errors are adjusted to account for serial correlation at the state level and heteroskedastic errors. All analyses use ACS survey weights

Dynamic DiD Event Study

To obtain unbiased and consistent causal estimates, the difference-in-differences (DiD) approach relies on the parallel trends assumption—that, in the absence of Medicaid expansion, outcomes for individuals in expansion and non-expansion states would have

followed similar trajectories. In practice, this assumption is often evaluated visually by comparing pre-expansion trends between treated and control groups. However, assessing pre-trends becomes more challenging under staggered adoption, where states implemented Medicaid expansion at different times.

To address this complexity and assess dynamic treatment effects, I extend the static TWFE model by estimating a dynamic DiD event-study specification. This model introduces a set of lead and lag indicators that capture changes in outcomes at multiple time points before and after a state adopts Medicaid expansion. This structure facilitates a more granular examination of whether the parallel trends assumption holds and reveals how treatment effects evolve over time.

Equation 2 outlines my preferred specification models, with Y_{ist} as the outcome of interest for individuals i in each year and state. To effectively account for the varied timing of Medicaid expansion, I employ lead and lag dummy variables denoted as (D^k_{its}). Each of these variables represents whether individual i in state s is observed in year t, which is exactly k years before or after Medicaid expansion took effect in their state. For example, D^{-2}_{its} equals 1 if the observation occurred two years before expansion and 0 otherwise; similarly, D^1_{its} equals 1 if the observation occurred one year after expansion. These variables allow the model to separately estimate how outcomes changed in each specific year leading up to and following expansion, rather than averaging the effect across all years. This helps us see not only whether the policy had an effect, but also when the effect began and how it changed over time. The omitted category is $k = -1$, corresponding to the year prior to expansion. Omitting the year $k = -1$ helps avoid issues of multicollinearity among the relative period indicators, as well as with the unit and time fixed effects.

The interaction term (FB_{ist} with leads and lags helps estimate the differential treatment effect for foreign-born individuals. The vectors X_{ist} and $\theta(FB_{ist} \cdot C_{ist})$, similar to Equation 1, include control variables and interactions that account for differing effects between foreign-born and US-born individuals. Additionally, (δ_s) and (λ_t) represent state

and year fixed effects.

$$Y_{its} = \sum_{k=-4}^{-2} \alpha_k D_{its}^k + FB_{ist} \sum_{k=-4}^{-2} \beta_k D_{its}^k + \sum_{k=0}^{5} \alpha_k D_{its}^k + FB_{ist} \sum_{k=0}^{5} \beta_k D_{its}^k + \theta(FB_{ist} \cdot C_{ist}) + \delta_s + \lambda_t + \psi X_{ist} + \epsilon_{its} \quad (2)$$

The estimated coefficients are visualized in event-study plots. If the parallel trends assumption holds, the coefficients on pre-treatment leads (α_k for $k < 0$ to not significantly differ from zero, while post-expansion coefficients should diverge only after the treatment year. All specifications apply survey weights and report robust standard errors clustered at the state level.

Addressing Challenges of Varying Treatment Effects Over Time

Since 2018, a growing body of literature, including notable studies by Callaway and Sant'Anna (2021), de Chaisemartin and D'Haultfœuille (2020), Goodman-Bacon (2021), Roth et al. (2023), and Sun and Abraham (2021), has raised concerns regarding the validation of staggered DiD models and event study designs. Goodman-Bacon showed that the estimated treatment effect in TWFE is the weighted average of all possible 2x2 DiD estimates. This includes groups that were treated early, groups that were treated later, and groups that were never treated. This theorem posits that if treatment effects vary over time, then this weighted average of all possible 2x2 DiD estimates provides biased estimates. A key concern arises when later-treated groups are compared with earlier-treated groups, whose treatment effects may have been growing over time. In certain situations, this can lead to a change in the sign of the parameter of interest, compromising the validity of inferences.

Likewise, Sun and Abraham (2021) underscored challenges with heterogeneous treatment and differential timing within event studies. They emphasized that effects from other time frames may also have an impact on the estimated effect associated with a particular lead or lag in addition to the actual treatment effect during that period. They further argued that apparent pre-trends prior to treatment could stem from treatment

effects varying across units (heterogeneity) rather than genuine pre-existing trends. To address these challenges, I compare the event study results with Sun and Abraham's (2021) Interaction-Weighted (IW) event study estimator, designed to be resilient against treatment effect heterogeneity across time and groups, assuming parallel trends and no anticipation behavior.

The IW estimator is derived by taking a weighted average of cohort-specific treatment effects using calculated weights from the sample shares of each cohort in the relevant periods. Specifically, it first estimates the Average Treatment Effect on the Treated for each cohort (CATT) using an interacted two-way fixed effects regression. Subsequently, group weights are estimated based on their sample shares during the relevant periods. Ultimately, the IW estimator is formulated by taking the weighted average of the estimated CATT in the first step and weight estimates in the second step. This methodology addresses inherent challenges with heterogeneous treatment effects, delivering more precise and interpretable estimates of treatment effects compared to conventional methods. To relax the parallel trends assumption and incorporate treatment timing dependence on covariates, Sun and Abraham (2021) advocate for techniques like the inverse propensity score reweighting and the doubly robust estimator proposed by Sant'Anna and Zhao (2020) as foundational elements of the IW estimator's final formulation.

To tackle concerns about the reliability of the TWFE estimator, I initially apply the Bacon decomposition to break down my primary estimates and examine the impact of heterogeneity on these estimates. Subsequently, I replicate my event study findings using the Sun and Abraham IW estimator, which yields consistent estimates despite the existence of treatment heterogeneity. I then compare its results with those of the event study.

Results

Descriptive Analysis

Table 1 functions as our baseline equivalence table, displaying weighted frequencies and means of characteristics, demographics, and insurance coverage for US-born and

foreign-born individuals in states with and without Medicaid expansion before the ACA enactment. This table facilitates comparisons within and across treatment (expansion) and control (non-expansion) groups. I conducted chi-squared tests to assess differences in categorical variables between groups. As can be seen in Table 1 a notable demographic variations exist between foreign-born and US-born adults, regardless of Medicaid expansion status. Differences are evident in education, employment, marital status, race, and disability status. For instance, in non-expansion states, foreign-born individuals are more likely to have less than a high school education, with a significant difference of 27%. They also have fewer disabilities (18%), a predominantly Hispanic racial composition (64%), and a 17% higher employment rate than their native counterparts. When comparing native-born or foreign-born individuals across expansion and non-expansion states, although the chi-squared test results show statistical significance, these differences are minimal, ranging between 0.15% to 3%. These small but significant differences are likely attributable to our large sample size.

Difference-in-Differences Model Results

Table 2 presents coefficient estimates from Specification 1, examining the effect of Medicaid expansion on our preferred outcomes, which include the uninsured rate, Medicaid coverage, employer-sponsored insurance, and directly purchased insurance. Each column represents a regression using a specific outcome variable. The linear probability model (LPM) was chosen for estimation over logit regression due to its superior fit in preliminary results and the ease of interpreting coefficients. All specifications account for heteroskedasticity-robust standard errors clustered at the state level and include state and year-fixed effects. The first row in the table displays the effect of expansion on the reference group, US-born; the second row displays the effect of being foreign-born on the probability of having different types of insurance; and the third row displays the difference in the effect of Medicaid expansion between foreign-born and US-born (Medicaid Expansion $\times$ Foreign-Born). All models include controls, state fixed effects, and year fixed

effects to account for additional factors influencing insurance coverage.

Column (1) suggests that, on average, Medicaid expansion decreases the probability of being uninsured for US-born individuals by 8%. Moreover, it reveals that foreign-born individuals are 3.3% less likely than US-born individuals to gain insurance.

Similarly, Columns (2) suggest that with the expansion of Medicaid eligibility, US-born individuals in expansion states were 14% more likely to gain Medicaid coverage, while eligible foreign-born individuals in the expansion states are 3.6% less likely than US-born individuals in expansion states to gain Medicaid.

Columns (3) and (4) show that, on average, Medicaid expansion reduces the probability of having private insurance, whether employer-sponsored or directly purchased, by about 2-3 percentage points. These effects slightly differ between foreign-born and US-born individuals. Foreign-born individuals are 1.7% more likely than US-born individuals to reduce their directly purchased insurance while not experiencing a significant change in employer-sponsored insurance.

Overall, the DiD results suggest that eligible US-born individuals are more likely to benefit from Medicaid expansion than eligible foreign-born individuals.

for corresponding results without controls and the results of different specifications of DID See Appendix A Tables 4 and 5

Result for DiD Event Study

To assess the impact of Medicaid expansion on uninsured rates among both US-born and foreign-born individuals, I employed an event-study model, as outlined in Equation 2.

The results of this specification is visually presented in Figure 1, with point estimates accompanied by 95% confidence intervals. Here, I estimated changes relative to the year preceding a state's adoption of the Medicaid expansion. Each panel in this figure displays the results for a given outcome variable. The black line with circles represents foreign-born individuals, and the purple line with triangles represents US-born individuals (for Unadjsuted Event study results see Appendix A figure 8).

Pre-trends:

For the uninsured outcome and private insurance (employer-based and directly purchased), pre-expansion coefficients are statistically insignificant and close to zero for both nativity groups, validating the parallel trends assumption. However, for Medicaid coverage, close to zero but statistically significant deviations are detected for foreign-born individuals, suggesting weaker support for the parallel trends assumption in that subgroup.

Post-treatment effects:

As illustrated in Figure 1, the expansion reduced uninsured rates for both groups, but the effect is larger and more sustained for U.S.-born adults. Medicaid coverage increased sharply post-expansion—by 16 percentage points for U.S.-born and approximately 12 percentage points for foreign-born adults.

Employer-sponsored coverage showed limited change. However, directly purchased insurance declined in both groups, especially among foreign-born adults—suggesting that Medicaid substituted for previously purchased coverage.

Subsample Analysis

Immigrant populations exhibit substantial diversity in legal status, socioeconomic factors, and integration levels—all of which mediate the effects of Medicaid expansion. While my analysis controls for observable characteristics (e.g., language proficiency, time in the U.S., citizenship), the lack of direct immigration-status data necessitated the use of a proxy for undocumentedness. This limitation is critical, as Medicaid eligibility varies sharply by legal status: documented immigrants often qualify after meeting residency requirements, whereas undocumented individuals face explicit exclusions outside emergency or state-funded programs.

To address this concern, I conducted additional analyses to examine the varying effects, employing a proxy for undocumentedness that I created using the Borjas residual methodology. Borjas (2017)introduced this novel proxy based on the Warren-Passel estimate, which is recognized as an 'official' estimate by the Department of Homeland

Security (DHS). Using a residual methodology, he applied a set of exclusion criteria to classify individuals as likely undocumented. According to these criteria, individuals are considered legal if they: arrive in the United States before 1980, were born in Cuba, hold U.S. citizenship, are current or former veterans or U.S. Armed Forces members, are recipients of public benefits , are government employees, hold a licensed occupation, or are married to a U.S. citizen. Those not meeting any of these criteria are classified as likely undocumented. While this updated proxy marks an improvement over previous methods and closely aligns with official DHS statistics, there remains a potential for overestimation and bias. Moreover, the DHS estimates themselves are not free from error or inherent bias.

Furthermore, ACS data has faced criticism for its underrepresentation of immigrant populations, particularly undocumented individuals. Due to fears associated with their unauthorized status, undocumented immigrants may be hesitant to participate in ACS surveys, leading to non-participation or misreporting of immigration status (Kissam, 2017; Warren, 2022). Additionally, the ACS employs a sampling strategy that randomly selects approximately 250,000 residential addresses per month, relying on the assumption that sampled households have stable and permanent residences (Dillingham et al., 2020). This "de facto" sampling method may systematically exclude immigrants who frequently move due to economic instability, legal pressures, or housing insecurity. Such populations are more likely to reside temporarily in shared housing arrangements, informal or non-standard housing, or otherwise lack a fixed residential address (Hall et al., 2019). Consequently, these conditions can lead to significant undercoverage of highly mobile immigrant populations (O'Hare, 2019). As a result, the ACS may not fully represent the lived experiences of many immigrants, particularly undocumented or transient individuals (Stempowski, 2023), thus introducing biases that underestimate the complexity and instability inherent in the immigrant experience.

Acknowledging this limitation, I estimated the impact of Medicaid expansion on insurance coverage among undocumented, legal immigrants in comparison with US-born.

The findings of this analysis are illustrated in Figure 2

As depicted in Figure 2, following Medicaid expansion, there was a decrease in uninsured rates and an increase in Medicaid coverage across all population groups. Additionally, a decline in privately purchased insurance is noticeable across the board for these groups.

Legal foreign-born and US-born individuals exhibit a similar trend in uninsured and Medicaid coverage. Notably, legal foreign-born individuals experienced a more significant decrease in directly purchased insurance, possibly shifting to the ACA's Insurance Marketplace. ACS did not collect any information on whether insurance was gained through the marketplace until 2018, which limits our ability to test this hypothesis.

Surprisingly, undocumented individuals, to a lesser extent, gained Medicaid benefits post-expansion in expansion states. This contrasts with their ineligibility for state-funded health coverage until 2020 (Kaiser Family Foundation, 2022). This anomaly raises a critical question: does Borjas' method combine undocumented individuals with low-income non-citizens (those low-income green card holders with no stable job or those on student or work visas or agricultural or seasonal workers)?

To delve deeper into this issue, I run the analysis using reported citizenship status available in the ACS data. The findings of this examination are presented in Figure 3.

The results show comparable trends for non-citizens and the undocumented proxy group, reinforcing the idea that the Borjas undocumented proxy measure may categorize many low-income non-citizen individuals as undocumented. While it is surprising to see take-up for undocumented individuals given the documentation requirement for Medicaid application. It is not surprising to see an uptake for Medicaid among non-citizens, as many are eligible for Medicaid expansion after meeting the five-year residency requirement.

Another possible explanation for the apparent increase in Medicaid coverage among undocumented immigrants could be that ACS responders might be misreporting their coverage type by misclassifying the emergency Medicaid benefits or charity care.

Together, these patterns suggest that caution is warranted when interpreting results for undocumented populations based on indirect proxies. Nonetheless, the findings underscore substantial within-immigrant heterogeneity in the effects of Medicaid expansion.

Given that a significant portion of the undocumented population in the U.S. is of Mexican origin, I conducted a complementary analysis restricted to individuals reporting Mexican nationality. Results are reported in Appendix A Figure 10. This subgroup shows similar coverage trends as the broader foreign-born population, but with slightly attenuated effects, consistent with known access barriers and immigration-related fears among Latino communities (Lambert et al., 2022).

Heterogeneity Analysis

Bacon decomposition

To address potential bias from staggered Medicaid expansion timing, I first employ Bacon decomposition (Goodman-Bacon, 2021), which reveals how treatment effect heterogeneity may distort traditional TWFE estimates. This method decomposes the TWFE estimator into a weighted average of three distinct 2×2 difference-in-differences (DiD) comparisons:

- Treated vs. Never-Treated: States that expanded Medicaid are compared to states that never expanded (the baseline "clean" comparison).
- Early-Treated vs. Late-Treated: States that expanded earlier (e.g., 2014 adopters) serve as controls for later-expanding states until the latter adopt the policy.
- Late-Treated vs. Early-Treated: Reverse of the above, where later-expanding states are compared to earlier adopters after the latter have already implemented expansion.

The primary concern with TWFE arises from the Late vs. Early-Treated comparisons. When treatment effects evolve over time (e.g., if early expansion states see larger initial impacts), these comparisons can introduce bias because later-treated units are compared to early-treated units that are no longer untreated controls.

Table 3 presents the primary estimate and the decomposed parameter estimate for three distinct types of 2x2 DiD. Panel A presents the results for the Uninsured, while Panel B details the decomposition related to Medicaid coverage. Figure 4 visually depicts this decomposition and its sources of variation. Each icon represents a 2x2 DiD, showcasing its weight and calculating the average effect, along with the total weight for the three 2x2 comparisons: treated/untreated, early/late, and late/early. The pink horizontal line denotes the estimated parameter from the TWFE. Similarly, Panel (a) illustrates the estimates for the Uninsured group, and Panel (b) displays the estimates for Medicaid coverage.

As demonstrated below, the main TWFE estimate is weighted average of these 2x2 estimates shown in the table 3

For Uninsured :

$$(-0.093098 * 0.11871) + (-0.05521 * 0.08771) + (-0.08462 * 0.79358) = -0.08304687$$

For Medicaid Coverage:

$$(0.14610 * 0.11871) + (0.08782 * 0.08771) + (0.14452 * 0.79358) = 0.1397344$$

As can be seen, the comparison between treated and untreated groups significantly influences the estimate. Whereas, the weights and the size of the problematic comparison group are relatively small. This observation aligns logically, particularly considering the limited number of late adopter states and the relatively brief post-period.

The DD estimate, excluding this particular variation source and focusing solely on comparisons between treated and untreated units, closely mirrors our primary DD estimate in magnitude. It also aligns more closely with the event-study coefficients. However, as discussed earlier, the event study specification is not without criticism, especially concerning heterogeneous treatment effects.

Sun & Abraham Interaction-weighted (IW) event study

To determine if the variation in treatment timing affects my event study results, I additionally estimated my model using the IW estimator Sun and Abraham (2021) and the Fixest package in R. Figure 5 displays the results from both the event study and the new IW event study estimator. In Panel (a), we observe results specific to the US-born population, while Panel (b) highlights findings for foreign-born individuals. The distinct darker line in both panels signifies estimates derived from the Sun & Abraham IW estimator, contrasting with the lighter lines that represent outcomes from our main event study.

Overall, the graph indicates striking similarities between the results of both the IW estimator and the event study model. Importantly, there remains a consistent absence of differential pre-trends. Based on this analysis, I conclude that my findings remain robust and have limited sensitivity to changes in treatment patterns across diverse groups.

Note that the current implementation of Sun & Abraham method exclusively supports difference-in-differences, rather than triple-differences. Consequently, in this context, I adhere to the specification outlined in Equation 3 in the appendix. Instead of utilizing an interaction term for foreign-born status, I opt for a split sample approach.

This careful consideration of treatment heterogeneity enhances the overall quality of our study and strengthens the validity of our results.

Discussion

This study aimed to estimate the impact of Medicaid expansion on insurance coverage among low-income adults, comparing foreign-born and US-born populations. Guided by theoretical and empirical insights into immigrant health access, I proceed to analyze using 2011-2019 ACS data. My findings indicate that Medicaid expansions increased both the likelihood of being insured and having Medicaid coverage for both groups. Moreover, I observed a significant but closing gap between foreign-born and US-born in overall insurance coverage and Medicaid coverage. Consistent in both event

study specification and DID fixed-effect model. These results are robust to Bacon decomposition and the Sun & Abraham interaction-weighted estimator, suggesting limited bias from heterogeneous treatment effects.

Subgroup analysis by citizenship status (Figure 3 and Table6) revealed that while Medicaid expansion improved coverage across all groups, substantial disparities persisted for foreign-born adults—particularly for non-citizens, who face explicit eligibility restrictions. U.S.-born adults experienced the largest gains: a +14.1 percentage point (pp) increase in Medicaid coverage and an -8.3 pp reduction in uninsured rates, with only modest crowd-out from other coverage sources. Foreign-born citizens also benefited, but their uninsured rate declined by a significantly smaller margin (+2.2 pp less reduction relative to U.S.-born adults), and they experienced greater crowd-out from direct-purchase insurance (-2.6 pp). Among foreign-born non-citizens, coverage gains were the smallest: their uninsured rate declined 4.0 pp less, and Medicaid coverage increased 6.2 pp less compared to U.S.-born adults Their crowd-out from directly purchased insurance was significantly but marginally higher than that of U.S.-born adults but lower than that of naturalized citizens. These contrasting "crowd-out" patterns underscore how eligibility and stigma interact. U.S.–born adults, who retained broad access to employer coverage and may attach significant social stigma to social assistance programs , replaced only a small share of directly purchased private insurance with Medicaid (Lundberg & Syltevik, 2024). Foreign-born adults' disproportionate shift from direct-purchase plans aligns with prior evidence suggesting that these populations face structural barriers to ESI—such as occupational segregation in sectors with limited benefits—which may increase reliance on non-group insurance (Buchmueller et al., 2007; Zallman et al., 2018). For naturalized citizens, lower out-of-pocket premiums in Medicaid than in the individual market and a weaker social stigma attached to public insurance made public coverage more attractive once legal risk seemed manageable. Roughly one-quarter of low-income foreign-born adults are undocumented or within five years of lawful entry; these non-citizens are categorically

barred by the 1996 PRWORA rule and are therefore not eligible for Medicaid (Straut-Eppsteiner, 2023). Those Medicaid-eligible non-citizens facing language hurdles, documentation requirements, and fears that enrollment could jeopardize future status (Joseph, 2016; Lambert et al., 2022) might forgo the benefit of Medicaid and stay uninsured or with their directly purchased insurance. A subset of non-citizens—such as those with Temporary Protected Status (TPS), student or work visas, or pending applications for lawful permanent residence—are eligible for Marketplace subsidies. These individuals can receive premium tax credits and other cost savings if their income is below 400% of the federal poverty level (FPL). However, because the ACS does not specifically identify Marketplace plans, respondents may report such coverage as either "direct-purchase" or even "Medicaid." As a result, among non-citizen foreign-born adults, gains in subsidized Marketplace coverage may offset reductions in direct-purchase insurance, leading to smaller observed declines. Some may even misreport subsidized Marketplace plans as Medicaid, further obscuring true transitions.

Moreover, as seen in the Figure3 the trajectory of Medicaid take-up among foreign-born citizens changed over time. In the early expansion years, naturalized citizens saw Medicaid gains similar to those of U.S.-born adults. However, a growing disparity emerged after 2017. This divergence aligns with a period of heightened restrictive immigration policies—including executive travel bans, increased interior enforcement, and public-charge rule changes. Applying Van Oorschot's multilevel model, such policies likely raised the perceived risk and uncertainty of applying for Medicaid, increasing non-price "application-stage" costs for naturalized citizens, even though they remained legally eligible for the program. This evolving disparity exemplifies how restrictive immigration measures can produce broader "chilling effects," discouraging benefit take-up even among populations not directly targeted (Barofsky et al., 2020; Lambert et al., 2022).

My results align with the existing body of research on the impact of Medicaid expansion among the foreign-born, such as those conducted by Bustamante et al. (2019),

Stimpson and Wilson (2018), and Tarraf et al. (2021). Confirming that expansion has effectively reduced uninsured rates among both foreign-born and U.S.-born individuals.

However, my study diverges from certain prior research regarding the closing of the coverage gap between U.S.-born and foreign-born individuals. While my results align with Stimpson and Wilson (2018) , indicating that Medicaid expansion did not eliminate this gap, they differ from the findings of Bustamante et al. (2018) The consistency with Stimpson's study can be attributed to the shared use of a similar methodology, namely the DiD approach, and reliance on the same dataset, ACS. Conversely, Bustamante's study, which yielded dissimilar results, was based on the NHIS and employed a comparison study design rather than causal inference methodology.

Furthermore, my research introduces a novel viewpoint to the existing body of research. Notably, to the best of my knowledge, no prior study has specifically examined the differential impact of Medicaid expansion on uninsured rates and Medicaid uptake between foreign-born and U.S.-born populations using a DiD event study controlling for the state political climate and state structural xenophobia score.

While this analysis offers valuable insights into the impact of Medicaid expansion on insurance take-up among foreign-born individuals, several limitations warrant consideration. First, our reliance on self-reported insurance status data from the ACS introduces potential measurement error. Respondents may inaccurately report their coverage, and the survey's design further complicates interpretation. For example, the ACS insurance questions did not distinguish subsidized Marketplace plans from other private coverage until 2018, grouping these under a vague category—"insurance purchased directly from an insurance company"—which risks conflating Marketplace, off-Marketplace, and other private plans. Similarly, respondents might misclassify subsidized plans as Medicaid, a concern supported by prior cognitive testing in Massachusetts (Boudreaux et al., 2019). Additionally, the ACS does not capture insurance churning (frequent coverage changes) or gaps in coverage over the year, which could distort findings. Another notable limitation is

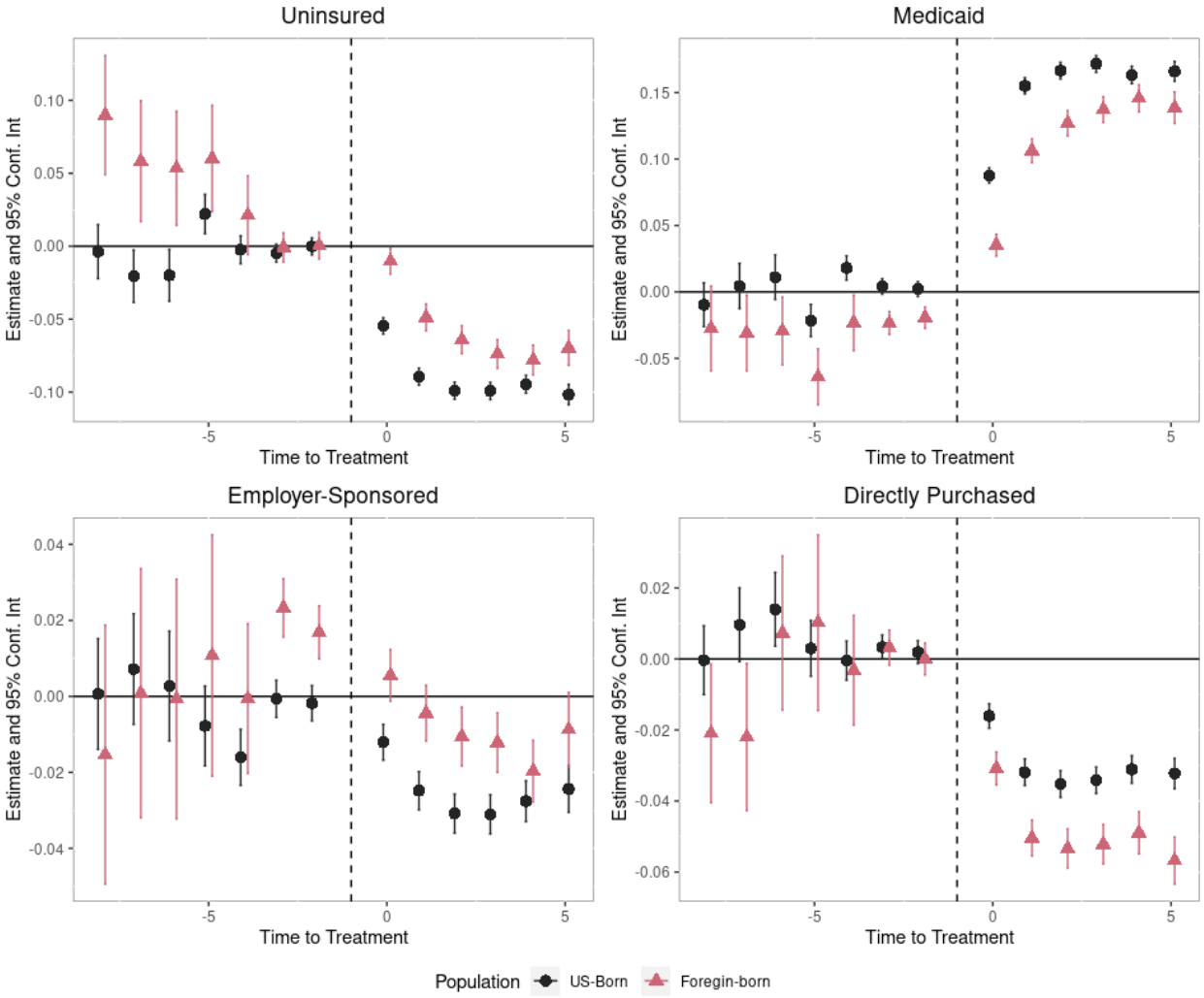
our inability to distinguish between legal and undocumented immigrants in our dataset. The exclusion of undocumented immigrants from Medicaid expansion eligibility, coupled with their lower survey participation rates, may affect the accuracy and completeness of presented results. Finally, this study excluded states such as New York, Massachusetts, Oregon, Vermont, and Delaware, which offered more generous health benefits prior to ACA expansion. Including these states could bias the estimated immigrant-native differences, as immigrants in those states may have already had greater access to Medicaid or state-funded coverage. As a result, the estimated immigrant impact of Medicaid expansion in a full sample may appear smaller, or the observed disparity between U.S.-born and foreign-born individuals may narrow. While this exclusion improves internal validity, it limits external generalizability to more inclusive policy contexts.

The study's findings emphasize the intricate landscape of healthcare policy implementation within diverse populations. While Medicaid expansion holds promise in mitigating uninsured rates and enhancing Medicaid coverage, the differential impact observed among foreign-born individuals underscores the imperative for a comprehensive approach. It is essential to acknowledge that achieving equitable healthcare access necessitates addressing not only coverage expansion but also dismantling structural, socioeconomic, and cultural barriers that may persist even in the presence of expanded healthcare programs. These findings also advocate for ongoing evaluation of policy impacts and underscore the importance of conducting research into the mechanisms by which Medicaid expansion influences access to care and ameliorates healthcare disparities among foreign-born and native populations. This avenue of research will provide valuable insights into the evolving dynamics of healthcare access, informing the development of more targeted and effective policies to ensure equitable healthcare for all.

Tables and Figures

Figure 1

DiD Event-study Result of the Effect Medicaid Expansions on Uninsured



Note: *This figure plots the coefficients from Specification 2*

Table 1

Baseline Characteristics (Weighted)

	Non-expansion			Expansion			Exp. Var.	
Characteristic	**FB**	**USB**	**Diff**	**FB**	**USB**	**Diff**	**FB**	**USB**
	N = 53,850	N = 248,302		N = 93,631	N = 326,186		N = 389,955	N = 1,585,639
Insurance Coverage, %								
Uninsured	67	40	28***	53	33	20***	14***	6.7***
Medicaid	14	30	-17***	27	38	-12***	-13***	-7.7***
Individually Purchased	5.0	8.3	-3.3***	5.4	8.8	-3.4***	-0.39**	-0.49**
Employer Sponsored	14	19	-4.7***	15	18	-2.8***	-1.6***	0.27**
Race, %								
Asian	8.2	0.2	8.1***	17	0.7	16***	-8.5***	-0.54***
Black	7.9	26	-19***	3.6	16	-13***	4.3***	10***
Hispanic	73	9.1	64***	66	9.7	57***	6.7***	-0.59***
Other	1.7	3.7	-2.0***	2.0	5.8	-3.8***	-0.27***	-2.1***
White	9.1	60	-51***	11	67	-56***	-2.2**	-6.9***
Sex, %								
Female	55	58	-3.1***	56	57	-1.4***	-0.32	1.4***
Male	45	42	3.1***	44	43	1.4***	0.32	-1.4
Employment Status, %								
Employed	55	38	17***	52	37	15***	2.9***	1.3***
Unemployed	9.8	14	-3.9***	12	15	-3.2***	-1.7***	-0.96***
Not in Labor Force	35	48	-13***	36	48	-12***	-1.2***	-0.34***
Education, %								
College Degree	7.0	7.3	-0.30*	7.2	8.3	-1.0***	-0.25	-0.96***
Graduate or Higher	2.9	2.4	0.48***	3.1	3.0	0.10	-0.22*	-0.60***
High School	25	37	-12***	23	37	-13***	1.6***	0.03
Less than High School	49	22	27***	50	19	31***	-1.3***	3.1***
Some College / Associate	16	31	-15***	16	33	-17***	0.15	-1.6***
Poverty, %								
Income 100–138% FPL	36	33	3.3***	37	32	5.0***	-0.66*	1.0***
Income <100% FPL	64	67	-3.3***	63	68	-5.0***	0.66*	-1.0***
Disability	11	29	-18***	11	28	-17***	0.07	0.91***
Married	60	34	26***	58	31	27***	1.4***	3.0***
Age, Mean (SD)	43 (10)	45 (11)	-2.4***	43 (10)	45 (11)	-1.8***	-0.44***	0.22***

Note. FB = Foreign-born; USB = U.S.-born; Exp. Var. = Expansion Variations. Weighted means for residents in expansion and non-expansion states pre-ACA by nativity status. Chi-squared tests use Rao & Scott's second-order correction; Wilcoxon rank-sum test for continuous variables. Significance: *** 0.01, ** 0.05, * 0.1.

Table 2

Impact of the ACA Expansions on Coverage, Controlled

	Uninsured	Medicaid	Employer-Sponsored	Directly Purchased
	[i]	[ii]	[iii]	[iv]
Medicaid Expansion	-0.083***	0.141***	-0.023***	-0.029***
	(0.002)	(0.002)	(0.001)	(0.001)
Foreign-Born	-0.083***	-0.038**	0.062***	0.031***
	(0.018)	(0.016)	(0.014)	(0.011)
Medicaid Expansion × Foreign-Born	0.033***	-0.036***	0.014***	-0.017***
	(0.003)	(0.003)	(0.002)	(0.002)
Mean of Dep. Var	0.45	0.29	0.22	0.07
Controls	yes	yes	yes	yes
State FE	yes	yes	yes	yes
Year FE	yes	yes	yes	yes
Observations	1,975,594	1,975,594	1,975,594	1,975,594
R^2	0.17791	0.19429	0.10807	0.05183
Within R^2	0.12404	0.13898	0.10586	0.04547

Notes: This table displays estimates for coefficients for the difference-in-differences model 1.

Heteroskedasticity-robust standard-errors in parentheses. Significance levels: ***: 0.01, **: 0.05, *: 0.1

Figure 2

DiD Event-Study Results by Nativity and Legal Status

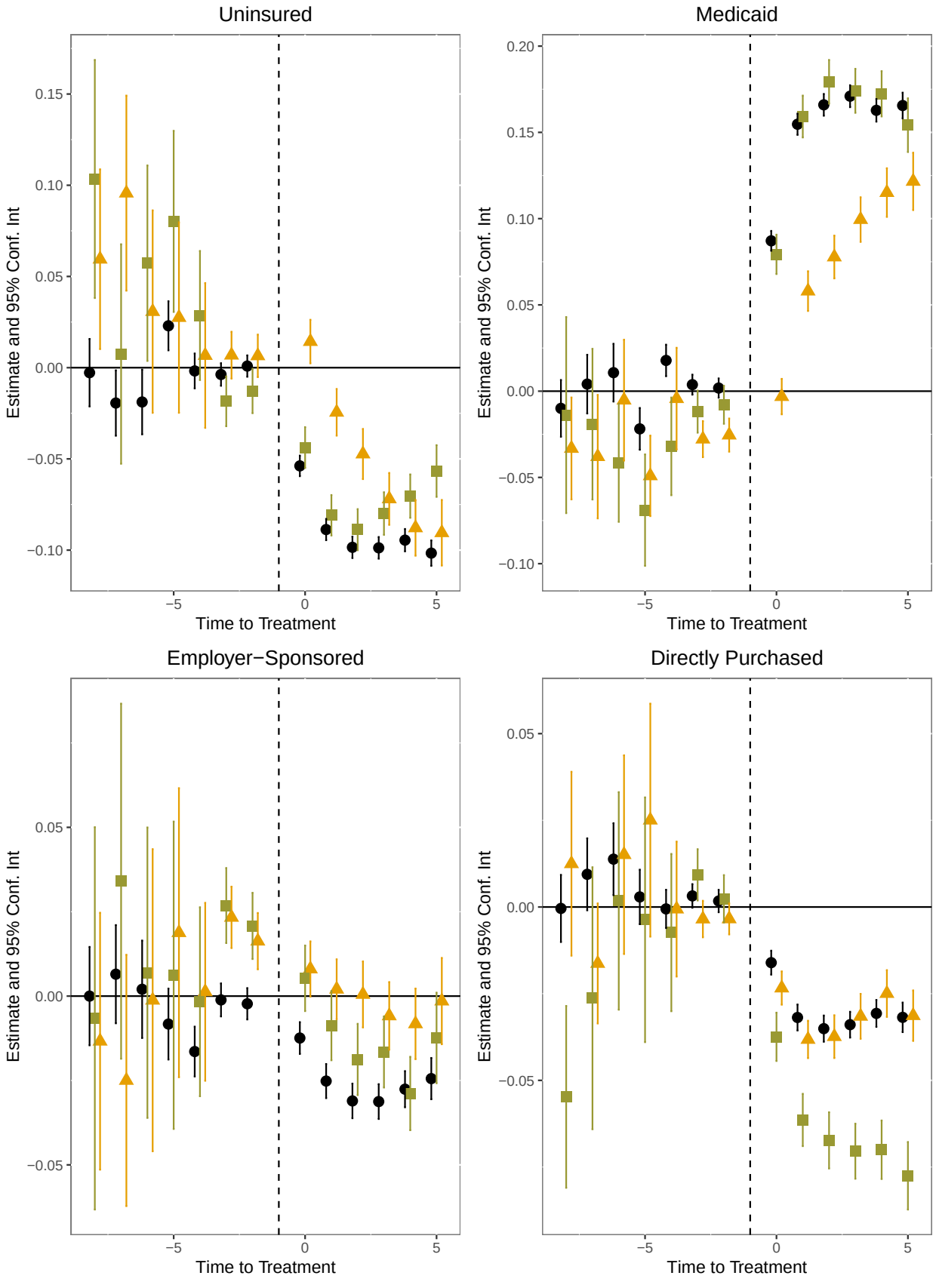


Figure 3

DiD Event-study Results by Nativity and Citizenship Status

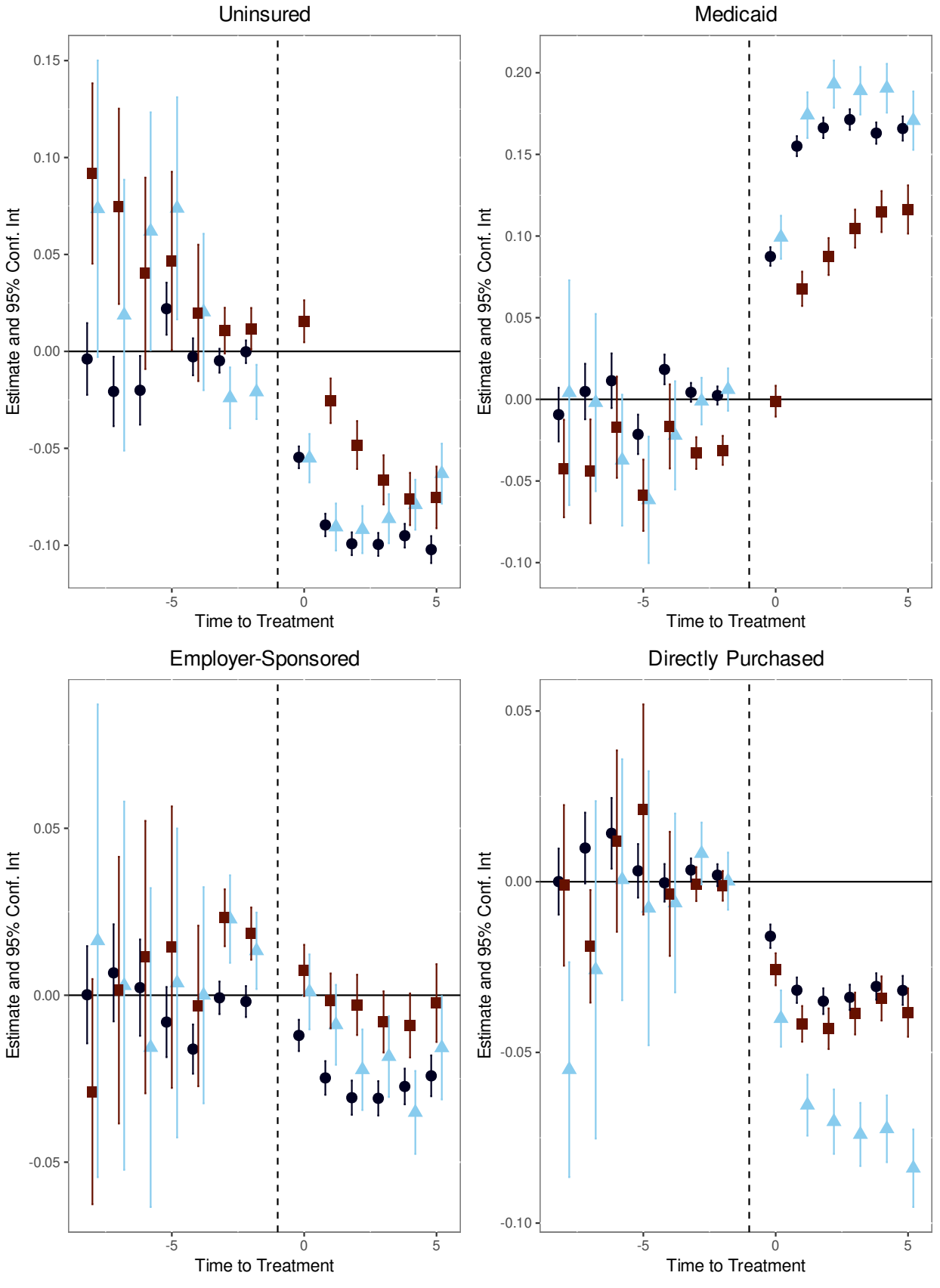


Figure 4

Bacon decomposition into weights and single 2x2 DiD

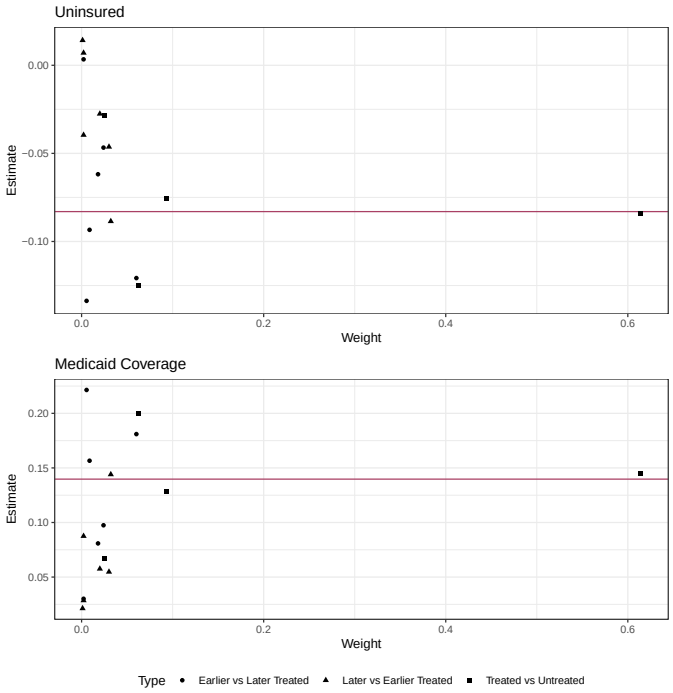


Table 3

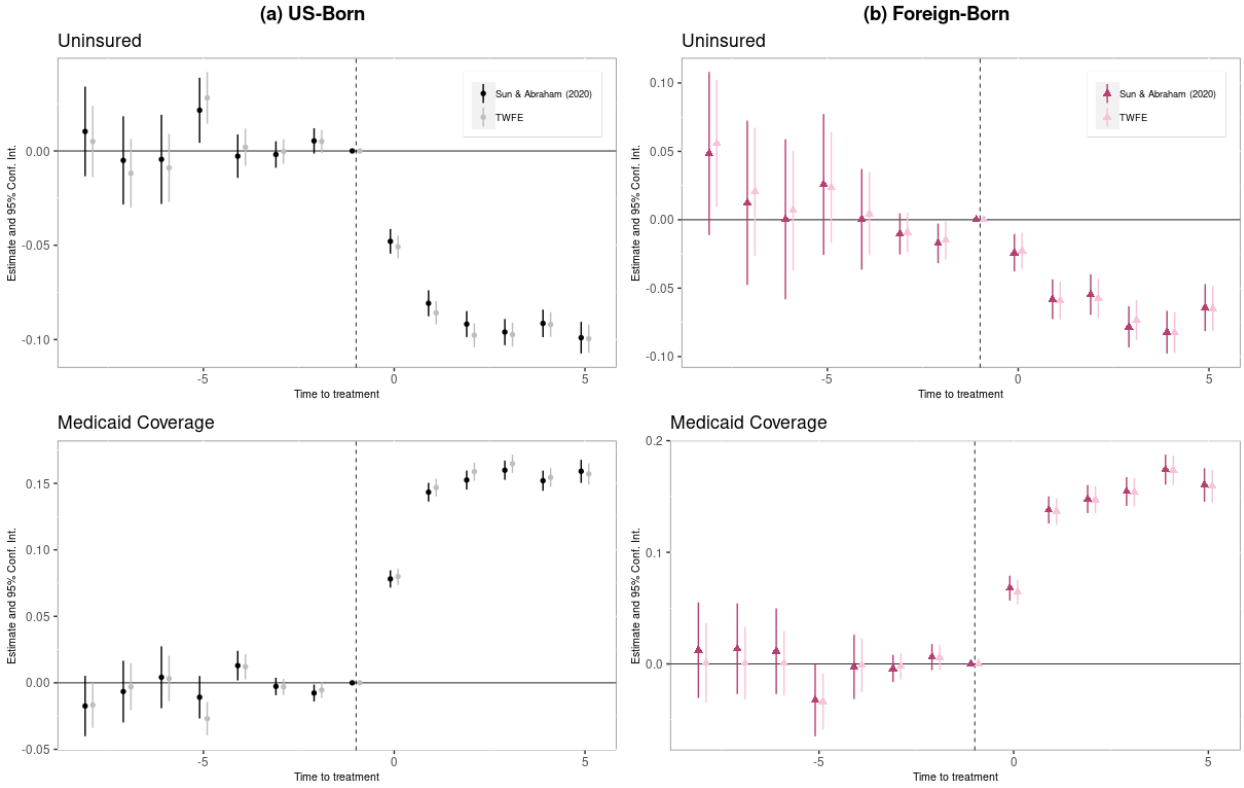
Bacon decomposition into weights and single 2x2 DiD

Comparison	Coefficient	Weight
Uninsured		
Earlier vs Later Treated	-0.09309	0.11871
Later vs Earlier Treated	-0.05521	0.08771
Treated vs Untreated	-0.08462	0.79358
Medicaid		
Earlier vs Later Treated	0.14610	0.11871
Later vs Earlier Treated	0.08782	0.08771
Treated vs Untreated	0.14452	0.79358

Notes: This table displays estimates for coefficients for the difference-in-differences model 1. Heteroskedasticity-robust standard-errors in parentheses. Significance levels: ***: 0.01, **: 0.05, *: 0.1

Figure 5

Effect of the ACA Medicaid Expansions Accounting for Variation in Treatment Timing Comparing Sun& Abraham estimator with Event study results



Note*: This figure reports the results using an alternative 'interaction-weighted' estimator from Sun and Abraham (2020) (lighter lines) and the event study result following Specification 3 (darker lines). Panel (a) plots coefficients for the US-born sample, and Panel (b) displays the coefficients for foreign-born sample.*

References

Adigun, S., Barroso, C., Mixer, S., Myers, C., & Anderson, J. (2021). Minding the Gaps: Health Care Access for Foreign-born People in the U.S.: An Integrative Review. *Journal of Health Care for the Poor and Underserved*, *32*(4), 1653–1674. https://doi.org/10.1353/hpu.2021.0158

Adunlin, G., Cyrus, J. W., Asare, M., & Sabik, L. M. (2019). Barriers and Facilitators to Breast and Cervical Cancer Screening Among Immigrants in the United States. *Journal of Immigrant and Minority Health*, *21*(3), 606–658. https://doi.org/10.1007/s10903-018-0794-6

Altman, C. E., Hamilton, C., Bachmeier, J. D., & Spence, C. (2023). Medicaid Health Insurance Coverage Among the Foreign-Born Following ACA Implementation: Disparities by Migration Status. *Population Research and Policy Review*, *42*(4), 68. https://doi.org/10.1007/s11113-023-09814-x

Angier, H. E., Marino, M., Springer, R. J., Schmidt, T. D., Huguet, N., & DeVoe, J. E. (2020). The Affordable Care Act improved health insurance coverage and cardiovascular-related screening rates for cancer survivors seen in community health centers. *Cancer*, *126*(14), 3303–3311. https://doi.org/10.1002/cncr.32900

Barofsky, J., Vargas, A., Rodriguez, D., & Barrows, A. (2020). Spreading Fear: The Announcement Of The Public Charge Rule Reduced Enrollment In Child Safety-Net Programs. *Health Affairs*, *39*(10), 1752–1761. https://doi.org/10.1377/hlthaff.2020.00763

Bird, E. J., Kayser, H., Frick, J. R., & Wagner, G. G. (1999, October). The Immigrant Welfare Effect: Take-Up or Eligibility? https://doi.org/10.2139/ssrn.199669

Borjas, G. J. (2017, March). The Earnings of Undocumented Immigrants. https://doi.org/10.3386/w23236

Borjas, G. J., & Trejo, S. J. (1991). Immigrant Participation in the Welfare System. *Industrial and Labor Relations Review*, *44*(2), 195–211. https://doi.org/10.2307/2524803

Boudreaux, M., Noon, J. M., Fried, B., & Pascale, J. (2019). Medicaid expansion and the Medicaid undercount in the American Community Survey. *Health Services Research*, *54*(6), 1263–1272. https://doi.org/10.1111/1475-6773.13213

Brodkin, E. Z., & Majmundar, M. (2010). Administrative Exclusion: Organizations and the Hidden Costs of Welfare Claiming. *Journal of Public Administration Research and Theory*, *20*(4), 827–848. https://doi.org/10.1093/jopart/mup046

Buchmueller, T. C., Levinson, Z. M., Levy, H. G., & Wolfe, B. L. (2016). Effect of the Affordable Care Act on Racial and Ethnic Disparities in Health Insurance Coverage. *American Journal of Public Health*, *106*(8), 1416–1421. https://doi.org/10.2105/AJPH.2016.303155

Buchmueller, T. C., Lo Sasso, A. T., Lurie, I., & Dolfin, S. (2007). Immigrants and Employer-Sponsored Health Insurance. *Health Services Research*, *42*(1p1), 286–310. https://doi.org/10.1111/j.1475-6773.2006.00600.x

Buettgens, M., Blavin, F., & Pan, C. (2021). The Affordable Care Act Reduced Income Inequality In The US. *Health Affairs (Project Hope)*, *40*(1), 121–129. https://doi.org/10.1377/hlthaff.2019.00931

Buettgens, M., & Ramchandani, U. (2023, May). *The Health Coverage of Noncitizens in the United States 2024* (tech. rep.). Urban Institute. Retrieved July 18, 2025, from https://www.urban.org/research/publication/health-coverage-noncitizens-united-states-2024

Bustamante, A. V., Chen, J., McKenna, R. M., & Ortega, A. N. (2019). Health Care Access and Utilization Among U.S. Immigrants Before and After the Affordable Care Act. *Journal of Immigrant and Minority Health*, *21*(2), 211–218. https://doi.org/10.1007/s10903-018-0741-6

Bustamante, A. V., McKenna, R. M., Viana, J., Ortega, A. N., & Chen, J. (2018). Access-To-Care Differences Between Mexican-Heritage And Other Latinos In California After The Affordable Care Act. *Health Affairs (Project Hope)*, *37*(9), 1400–1408. https://doi.org/10.1377/hlthaff.2018.0416

Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-Differences with multiple time periods. *Journal of Econometrics*, *225*(2), 200–230. https://doi.org/10.1016/j.jeconom.2020.12.001

Carrasquillo, O., Carrasquillo, A. I., & Shea, S. (2000). Health insurance coverage of immigrants living in the United States: Differences by citizenship status and country of origin. *American Journal of Public Health*, *90*(6), 917–923. Retrieved April 23, 2022, from https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1446276/

Chang, C. D. (2019). Social Determinants of Health and Health Disparities Among Immigrants and their Children. *Current Problems in Pediatric and Adolescent Health Care*, *49*(1), 23–30. https://doi.org/10.1016/j.cppeds.2018.11.009

Choi, S., Kim, G., & Lee, S. (2016). Effects of Nativity, Length of Residence, and County-Level Foreign-Born Density on Mental Health Among Older Adults in the U.S. *Psychiatric Quarterly*, *87*(4), 675–688. https://doi.org/10.1007/s11126-016-9418-2

Chu, J., Ortega, A. N., Park, S., Vargas-Bustamante, A., & Roby, D. H. (2021). The Affordable Care Act and Health Care Access and Utilization Among White, Asian, and Latino Immigrants in California. *Medical Care*, *59*(9), 762–767. https://doi.org/10.1097/MLR.0000000000001586

Clarke, A., & Isphording, I. E. (2017). Language Barriers and Immigrant Health. *Health Economics*, *26*(6), 765–778. https://doi.org/10.1002/hec.3358

Courtemanche, C., Marton, J., Ukert, B., Yelowitz, A., & Zapata, D. (2017). Early Impacts of the Affordable Care Act on Health Insurance Coverage in Medicaid Expansion

and Non-Expansion States. *Journal of Policy Analysis and Management*, *36*(1), 178–210. https://doi.org/10.1002/pam.21961

Courtemanche, C., Marton, J., Ukert, B., Yelowitz, A., & Zapata, D. (2018). Effects of the Affordable Care Act on Health Care Access and Self-Assessed Health After 3 Years. *Inquiry: A Journal of Medical Care Organization, Provision and Financing*, *55*, 46958018796361. https://doi.org/10.1177/0046958018796361

Courtemanche, C. J., Jones, J. W., Koumpias, A. M., & Zapata, D. (2023, January). Revisiting the Connection Between State Medicaid Expansions and Adult Mortality. https://doi.org/10.3386/w30818

De, P. K. (2021). Impacts of insurance expansion on health cost, health access, and health behaviors: Evidence from the medicaid expansion in the US. *International Journal of Health Economics and Management*, *21*(4), 495–510. https://doi.org/10.1007/s10754-021-09306-5

de Chaisemartin, C., & D'Haultfœuille, X. (2020). Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects. *American Economic Review*, *110*(9), 2964–2996. https://doi.org/10.1257/aer.20181169

Dillingham, S., Jarmin, R., Fontenot, A. E., & Stempowski, D. M. (2020). Understanding and Using American Community Survey Data: What All Data Users Need to Know.

Furtado, D., & Theodoropoulos, N. (2016). IMMIGRANT NETWORKS AND THE TAKE-UP OF DISABILITY PROGRAMS: EVIDENCE FROM U.S. CENSUS DATA. *Economic Inquiry*, *54*(1), 247–267. https://doi.org/10.1111/ecin.12248

Gee, E. R., & Giuntella, G. O. (2011). Medicaid and Ethnic Networks. *The B.E. Journal of Economic Analysis & Policy*, *11*(1). https://doi.org/10.2202/1935-1682.2990

Giulietti, C. (2014). The welfare magnet hypothesis and the welfare take-up of migrants. *IZA World of Labor*. https://doi.org/10.15185/izawol.37

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, *225*(2), 254–277. https://doi.org/10.1016/j.jeconom.2021.03.014

Gooptu, A., Moriya, A. S., Simon, K. I., & Sommers, B. D. (2016). Medicaid Expansion Did Not Result In Significant Employment Changes Or Job Reductions In 2014. *Health Affairs*, *35*(1), 111–118. https://doi.org/10.1377/hlthaff.2015.0747

Gutierrez, J. R. (2021). Healthy Communities: Shifting Toward Inclusive Immigration Policy. *Pediatrics*, *147*(2), e2020045880. https://doi.org/10.1542/peds.2020-045880

Hall, M., Musick, K., & Yi, Y. (2019). Living Arrangements and Household Complexity among Undocumented Immigrants. *Population and Development Review*, *45*(1), 81–101. Retrieved April 22, 2025, from https://www.jstor.org/stable/45174477

Hoynes, H. W., Page, M. E., & Stevens, A. H. (2006). Poverty in America: Trends and Explanations. *Journal of Economic Perspective*, *20*(1), 47–68.

Janssens, J., & Van Mechelen, N. (2022). To take or not to take? An overview of the factors contributing to the non-take-up of public provisions. *European Journal of Social Security*, *24*(2), 95–116. https://doi.org/10.1177/13882627221106800

Joseph, T. D. (2016). What Health Care Reform Means for Immigrants: Comparing the Affordable Care Act and Massachusetts Health Reforms. *Journal of Health Politics, Policy and Law*, *41*(1), 101–116. https://doi.org/10.1215/03616878-3445632

Kaiser Family Foundation. (2022, April). Status of State Medicaid Expansion Decisions: Interactive Map. Retrieved April 22, 2022, from https://www.kff.org/medicaid/issue-brief/status-of-state-medicaid-expansion-decisions-interactive-map/

Kerr, S. A. (1982). Deciding about Supplementary Pensions: A Provisional Model. *Journal of Social Policy*, *11*(4), 505–517. https://doi.org/10.1017/S0047279400022558

Khullar, D., & Chokshi, D. A. (2019). Challenges for immigrant health in the USA—the road to crisis. *The Lancet*, *393*(10186), 2168–2174. https://doi.org/10.1016/S0140-6736(19)30035-2

Kissam, E. (2017). Differential undercount of Mexican immigrant families in the U.S. Census. *Statistical Journal of the IAOS*, *33*(3), 797–816. https://doi.org/10.3233/SJI-170388

Kline, N. (2019). When deservingness policies converge: US immigration enforcement, health reform and patient dumping. *Anthropology & Medicine*, *26*(3), 280–295. https://doi.org/10.1080/13648470.2018.1507101

Ko, W., & Moffitt, R. A. (2022). Take-up of Social Benefits. *National Bureau of Economic Research.*, (w30148), 60.

Ku, L., & Bruen, B. (2013). The Use of Public Assistance Benefits by Citizens and Non-Citizen Immigrants in the United States. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.2228764

Lambert, R. F., Discepolo, K. E., & Elani, H. W. (2022). Chilling Effects on Immigrants' Health Insurance Coverage After the 2016 Presidential Election. *Journal of Immigrant and Minority Health*, *24*(4), 819–826. https://doi.org/10.1007/s10903-022-01349-1

Lee, H., & Porell, F. W. (2020). The Effect of the Affordable Care Act Medicaid Expansion on Disparities in Access to Care and Health Status. *Medical Care Research and Review*, *77*(5), 461–473. https://doi.org/10.1177/1077558718808709

Li, Y., Toseef, M. U., Jensen, G. A., Ortiz, K., González, H. M., & Tarraf, W. (2021). Gains in insurance coverage following the affordable care act and change in preventive services use among non-elderly US immigrants. *Preventive Medicine*, *148*, 106546. https://doi.org/10.1016/j.ypmed.2021.106546

Lin, Y., Monnette, A., & Shi, L. (2021). Effects of medicaid expansion on poverty disparities in health insurance coverage. *International Journal for Equity in Health*, *20*, 171. https://doi.org/10.1186/s12939-021-01486-3

Lipsky, M. (2010, April). *Street-Level Bureaucracy, 30th Anniversary Edition: Dilemmas of the Individual in Public Service*. Russell Sage Foundation.

Lipton, B. J., Decker, S. L., & Sommers, B. D. (2019). The Affordable Care Act Appears to Have Narrowed Racial and Ethnic Disparities in Insurance Coverage and Access to Care Among Young Adults. *Medical Care Research and Review*, *76*(1), 32–55. https://doi.org/10.1177/1077558717706575

Lucas, B., Bonvin, J.-M., & Hümbelin, O. (2021). The Non-Take-Up of Health and Social Benefits: What Implications for Social Citizenship? *Swiss Journal of Sociology*, *47*(2), 161–180. https://doi.org/10.2478/sjs-2021-0013

Lundberg, K. G., & Syltevik, L. J. (2024). Welfare stigma in a social democratic welfare regime during a decade of national public debate: Production, contestation and continuities. *European Societies*, *26*(5), 1572–1595. https://doi.org/10.1080/14616696.2024.2333526

Mallet, M., & Garcia, E. (2021). Caseworker prejudice: Exploring secondary non-take-up from below among Latino immigrants in Madrid, London, and New York City. *Swiss Journal of Sociology*, *47*(2). Retrieved September 20, 2022, from https://ora.ox.ac.uk/objects/uuid:54917ad8-d236-480b-be09-8c12f39fae7c

Marrow, H. B., & Joseph, T. D. (2015). Excluded and Frozen Out: Unauthorised Immigrants' (Non)Access to Care after US Health Care Reform. *Journal of Ethnic and Migration Studies*, *41*(14), 2253–2273. https://doi.org/10.1080/1369183X.2015.1051465

Mazurenko, O., Balio, C. P., Agarwal, R., Carroll, A. E., & Menachemi, N. (2018). The Effects Of Medicaid Expansion Under The ACA: A Systematic Review. *Health Affairs*, *37*(6), 944–950. https://doi.org/10.1377/hlthaff.2017.1491

Miller, S., Johnson, N., & Wherry, L. R. (2021). Medicaid and Mortality: New Evidence From Linked Survey and Administrative Data. *The Quarterly Journal of Economics*, *136*(3), 1783–1829. https://doi.org/10.1093/qje/qjab004

Miller, S., & Wherry, L. R. (2017). Health and Access to Care during the First 2 Years of the ACA Medicaid Expansions. *New England Journal of Medicine*, *376*(10), 947–956. https://doi.org/10.1056/NEJMsa1612890

Nyman, J. A. (2003). *The Theory of Demand for Health Insurance*. Stanford University Press.

O'Hare, W. P. (2019). Who Is Missing? Undercounts and Omissions in the U.S. Census. In W. P. O'Hare (Ed.), *Differential Undercounts in the U.S. Census: Who is Missed?* (pp. 1–12). Springer International Publishing. https://doi.org/10.1007/978-3-030-10973-8_1

Ortega, A. N., McKenna, R. M., Chen, J., Alcalá, H. E., Langellier, B. A., & Roby, D. H. (2018). Insurance Coverage and Well-Child Visits Improved for Youth Under the Affordable Care Act, but Latino Youth Still Lag Behind. *Academic Pediatrics*, *18*(1), 35–42. https://doi.org/10.1016/j.acap.2017.07.006

Pandey, S. K., Cantor, J. C., & Lloyd, K. (2014). Immigrant Health Care Access and the Affordable Care Act. *Public Administration Review*, *74*(6), 749–759. https://doi.org/10.1111/puar.12280

Park, S., Stimpson, J. P., Pintor, J. K., Roby, D. H., McKenna, R. M., Chen, J., & Ortega, A. N. (2019). The Effects of the Affordable Care Act on Health Care Access and Utilization Among Asian American Subgroups. *Medical Care*, *57*(11), 861–868. https://doi.org/10.1097/MLR.0000000000001202

Pauly, M. V. (1968). The Economics of Moral Hazard: Comment. *The American Economic Review*, *58*(3), 531–537. Retrieved May 15, 2022, from http://www.jstor.org/stable/1813785

Perreira, K. M., & Pedroza, J. M. (2019). Policies of Exclusion: Implications for the Health of Immigrants and Their Children. *Annual Review of Public Health*, *40*, 147–166. https://doi.org/10.1146/annurev-publhealth-040218-044115

Porteny, T., Ponce, N., & Sommers, B. D. (2022). Immigrants and the Affordable Care Act: Changes in Coverage and Access to Care by Documentation Status. *Journal of Immigrant and Minority Health*, *24*(1), 86–94. https://doi.org/10.1007/s10903-020-01124-0

Roth, J., Sant'Anna, P. H., Bilinski, A., & Poe, J. (2023). What's trending in difference-in-differences? A synthesis of the recent econometrics literature. *Journal of Econometrics*, S0304407623001318. https://doi.org/10.1016/j.jeconom.2023.03.008

Ryan, L., Sales, R., Tilki, M., & Siara, B. (2008). Social Networks, Social Support and Social Capital: The Experiences of Recent Polish Migrants in London. *Sociology*, *42*(4), 672–690. https://doi.org/10.1177/0038038508091622

Samari, G., Nagle, A., & Coleman-Minahan, K. (2021). Measuring structural xenophobia: US State immigration policy climates over ten years. *SSM - Population Health*, *16*, 100938. https://doi.org/10.1016/j.ssmph.2021.100938

Sant'Anna, P. H. C., & Zhao, J. (2020). Doubly robust difference-in-differences estimators. *Journal of Econometrics*, *219*(1), 101–122. https://doi.org/10.1016/j.jeconom.2020.06.003

Schumacher, S., Presiado, M., & Gonzalez-Barrera, A. (2023). *Understanding the U.S. Immigrant Experience: The 2023 KFF/LA Times Survey of Immigrants* (tech. rep.). Kaiser Family Foundation. Retrieved September 18, 2023, from https://www.kff.org/racial-equity-and-health-policy/poll-finding/kff-la-times-survey-of-immigrants/

Skinner, C. (2012). State Immigration Legislation and SNAP Take-Up Among Immigrant Families with Children. *Journal of Economic Issues*, *46*(3), 661–682. https://doi.org/10.2753/JEI0021-3624460304

Soni, A., Hendryx, M., & Simon, K. (2017). Medicaid Expansion Under the Affordable Care Act and Insurance Coverage in Rural and Urban Areas. *The Journal of Rural Health*, *33*(2), 217–226. https://doi.org/10.1111/jrh.12234

Soni, A., Wherry, L. R., & Simon, K. I. (2020). How Have ACA Insurance Expansions Affected Health Outcomes? Findings From The Literature. *Health Affairs*, *39*(3), 371–378. https://doi.org/10.1377/hlthaff.2019.01436

State's Health Fact. (2013, April). About State Health Facts. Retrieved September 24, 2022, from https://www.kff.org/about-state-health-facts/

Stempowski, D. (2023, November). Counting Every Voice: Understanding Hard-to-Count and Historically Undercounted Populations. Retrieved April 22, 2025, from https://www.census.gov/newsroom/blogs/random-samplings/2023/10/understanding-undercounted-populations.html

Stimpson, J. P., & Wilson, F. A. (2018). Medicaid Expansion Improved Health Insurance Coverage For Immigrants, But Disparities Persist. *Health Affairs*, *37*(10), 1656–1662. https://doi.org/10.1377/hlthaff.2018.0181

Straut-Eppsteiner, H. (2023). *Citizenship and Immigration Statuses of the U.S. Foreign-Born Population* (tech. rep. No. IF11806). Congressional l Research Service.

Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, *225*(2), 175–199. https://doi.org/10.1016/j.jeconom.2020.09.006

Tarraf, W., Jensen, G. A., Li, Y., Toseef, M. U., Mahmoudi, E., & Gonzalez, H. M. (2021). Changes in Insurance Coverage and Healthcare Use Among Immigrants and US-Born Adults Following the Affordable Care Act. *Journal of Racial and Ethnic Health Disparities*, *8*(2), 363–374. https://doi.org/10.1007/s40615-020-00790-y

Tolbert, J., Cervantes, S., Bell, C., & Published, A. D. (2024, December). Key Facts about the Uninsured Population. Retrieved April 21, 2025, from https://www.kff.org/uninsured/issue-brief/key-facts-about-the-uninsured-population/

Toyoda, Y., Oh, E. J., Premaratne, I. D., Chiuzan, C., & Rohde, C. H. (2020). Affordable Care Act State-Specific Medicaid Expansion: Impact on Health Insurance Coverage and Breast Cancer Screening Rate. *Journal of the American College of Surgeons*, *230*(5), 775–783. https://doi.org/10.1016/j.jamcollsurg.2020.01.031

Tumlin, K. C., & Zimmerman, W. (2003, October). *Immigrants and TANF: A Look at Immigrant Welfare Recipients in Three Cities. Occasional Paper. Assessing the New Federalism: An Urban Institute Program to Assess Changing Social Policies*. Urban Institute, 2100 M Street, N. Retrieved September 21, 2022, from https://eric.ed.gov/?id=ED482416

US Census Bureau. (2023). Response Rates. Retrieved May 16, 2022, from https://www.census.gov/programs-surveys/acs/

van Oorschot, W. (1998). Failing selectivity: On the extent and causes of non-take-up of social security benefits. In H.-J. Andress (Ed.), *Empirical poverty research in a comparative perspective* (pp. 101–132). Aldershot.

Warren, R. (2022). 2020 American Community Survey: Use with Caution, An Analysis of the Undercount in the 2020 ACS Data Used to Derive Estimates of the Undocumented Population. *Journal on Migration and Human Security*, *10*(2), 134–145. https://doi.org/10.1177/23315024221102327

Wehby, G. L., & Lyu, W. (2018). The Impact of the ACA Medicaid Expansions on Health Insurance Coverage through 2015 and Coverage Disparities by Age, Race/Ethnicity, and Gender. *Health Services Research*, *53*(2), 1248–1271. https://doi.org/10.1111/1475-6773.12711

Wen, H., Hockenberry, J. M., & Cummings, J. R. (2017). The effect of Medicaid expansion on crime reduction: Evidence from HIFA-waiver expansions. *Journal of Public Economics*, *154*, 67–94. https://doi.org/10.1016/j.jpubeco.2017.09.001

Yue, D., Rasmussen, P. W., & Ponce, N. A. (2018). Racial/Ethnic Differential Effects of Medicaid Expansion on Health Care Access. *Health Services Research*, *53*(5), 3640–3656. https://doi.org/10.1111/1475-6773.12834

Zallman, L., Woolhandler, S., Touw, S., Himmelstein, D. U., & Finnegan, K. E. (2018). Immigrants Pay More In Private Insurance Premiums Than They Receive In Benefits. *Health Affairs*, *37*(10), 1663–1668. https://doi.org/10.1377/hlthaff.2018.0309

Zerhouni, Y. A., Trinh, Q.-D., Lipsitz, S., Goldberg, J., Irani, J., Bleday, R., Haider, A. H., & Melnitchouk, N. (2019). Effect of Medicaid Expansion on Colorectal Cancer Screening Rates. *Diseases of the Colon & Rectum*, *62*(1), 97–103. https://doi.org/10.1097/DCR.0000000000001260

Causal Structure Discovery

I utilized causal structure discovery techniques to construct a comprehensive Directed Acyclic Graph (DAG) directly from the data, employing algorithms such as GDS, GES, PC, and FCI. The final DAG was determined through a majority voting approach following Joe et al. (2023). Subsequent adjustments were made to align the graph with real-world expertise, simplifying intricate relationships. This derived DAG not only facilitated hypothesis generation but also played a crucial role in guiding the analysis process. Additionally, it aided in the selection of pertinent control variables based on the identified relationships, ensuring a robust evaluation of Medicaid expansion effects while accounting for variable interactions and potential biases.

The DAG Graph depicted in 6

Estimates Event Study, Split Sample Based On Nativity

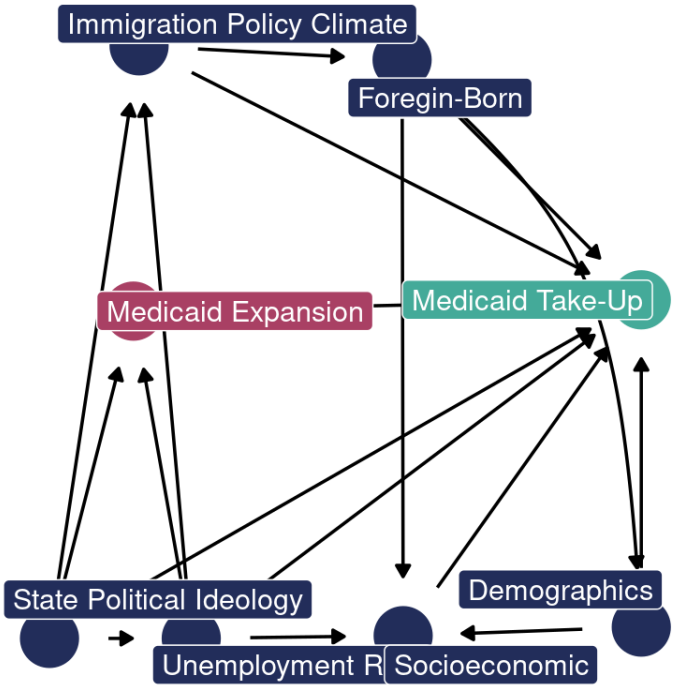
In the main analysis for the event study, I used the whole sample and employed the foreign-born status to evaluate the difference between the foreign-born and US-born. Here I will use the following specification and estimate the effect of Medicaid expansion separately for foreign-born and US-born.

$$Y_{its} = \alpha + \delta_s + \lambda_t + \psi X_{ist} + \sum_{k=-4}^{-2} \beta_k D_{its}^k + \sum_{k=0}^{5} \beta_k D_{its}^k + \epsilon_{its} \quad (3)$$

In this equation, Y_{ist} represents the outcome of interest for an individual i in year t in state s, δ_s captures state-fixed effects, and λ_t accounts for year-fixed effects. (D_{its}^k) represent a series of lead and lags. These dummy variables measure the time relative to the implementation year, enabling the assessment of changes in outcomes within expansion states compared to non-expansion states during the specific year (k), with the parameter (β_k) representing the estimated effect. The omitted category is $k = -1$, corresponding to the year prior to expansion. Omitting the year $k = -1$ helps avoid issues of multicollinearity among the relative period indicators, as well as with the unit and time fixed effects. X_{ist} is a vector series comprising individual demographic and socioeconomic characteristics, disability status, and time-varying controls for state-level economic and

Figure 6

The DAG From causal structure discovery

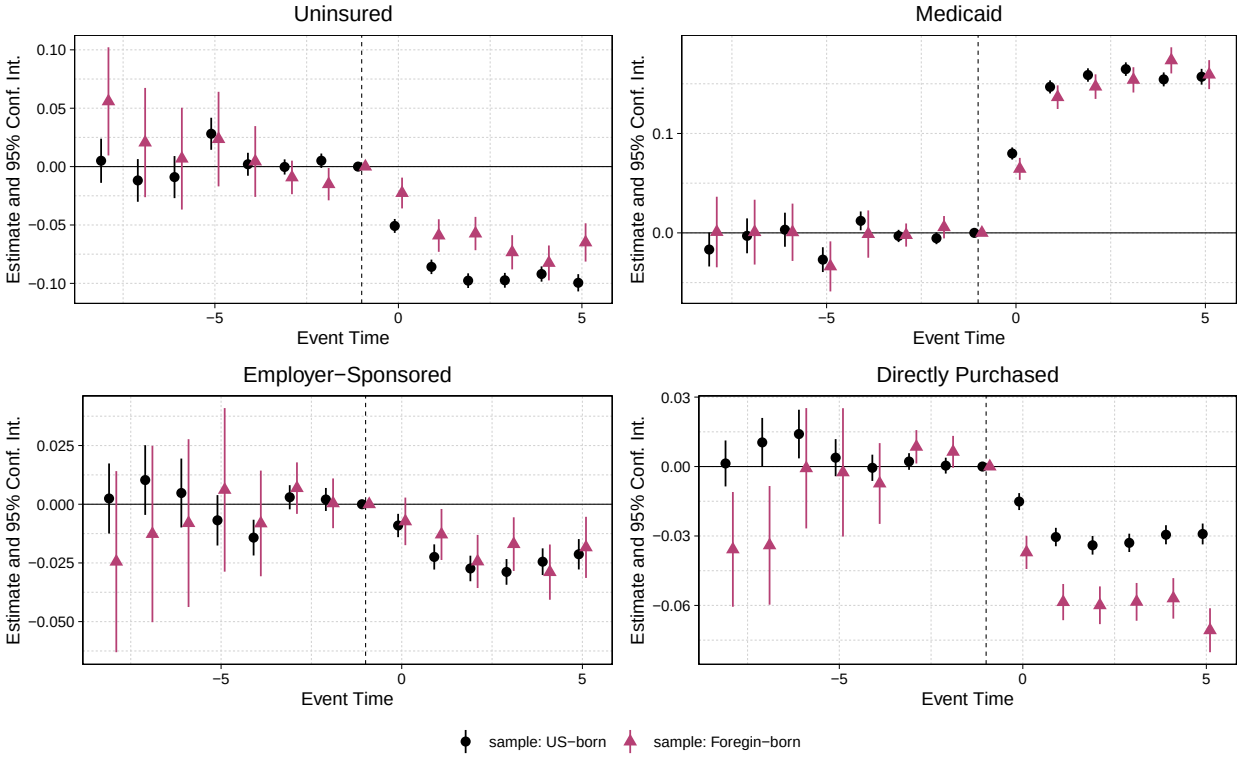


political characteristics, as detailed in the framework developed in Section 2. For the US-born sample, the X vector includes age, sex, disability status, poverty ratio, education level, marital status, race and ethnicity, state unemployment rate, and state measure of political liberalism. For the foreign sample, additional variables such as English proficiency, percentage of life lived in the U.S., non-citizenship, and state structural xenophobia are added to the covariates.

The result of this specification is depicted in figure 7 tables 7 and 8. To see the Event study result with no control check figure 9

Figure 7

Event-study regression controlled split sample



Note: This figure plots the coefficients from Specification 3

Table 4

Descriptive Statistics of Main Sample by State Expansion Status

Characteristic	Nonexpansion $n = 152,664,719$	Expansion $n = 59,137,774$	p-value[2]
Disability (%)	41,726,325 (27%)	116,770,669 (28%)	< 0.001
Sex (%)			< 0.001
Female	88,070,872 (58%)	33,786,721 (57%)	
Male	64,593,847 (42%)	25,351,053 (43%)	
Married (%)	44,677,404 (29%)	15,294,163 (26%)	< 0.001
Education Level (%)			< 0.001
Less than High School	32,256,888 (21%)	10,876,530 (18%)	
High School	56,194,673 (37%)	21,566,219 (36%)	
Some college or Associate degree	48,486,201 (32%)	19,557,798 (33%)	
College Degree	11,790,859 (7.7%)	5,239,393 (8.9%)	
Graduate and Beyond	3,936,098 (2.6%)	1,897,834 (3.2%)	
Employment Status (%)			< 0.001
Employed	60,151,014 (39%)	23,231,100 (39%)	
Not in Labor Force	72,377,154 (47%)	29,901,409 (51%)	
Unemployed	20,136,551 (13%)	6,005,265 (10%)	
Federal Poverty Level (%)			< 0.001
Between 100-138% FPL	49,215,196 (32%)	17,956,823 (30%)	
Below 100% of FPL	103,449,523 (68%)	41,180,951 (70%)	
Hispanic (%)	16,751,692 (11%)	8,046,108 (14%)	< 0.001
Race (%)			< 0.001
White	103,252,245 (68%)	41,348,962 (70%)	
Asian	790,874 (0.5%)	704,493 (1.2%)	
Black	38,496,717 (25%)	11,229,840 (19%)	
American Indian or Alaska Native	2,618,072 (1.7%)	1,317,643 (2.2%)	
Other	7,506,811 (4.9%)	4,536,836 (7.7%)	
Mean Age (yr)	43 (33, 54)	43 (34, 55)	< 0.001

Table 5

Impact of the ACA Expansions on Coverage Uncontrolled (Difference-in-differences Model)

	Uninsured	Medicaid	Employer Sponsored	Directly Purchased
	[i]	[ii]	[iii]	[iv]
Medicaid Expansion	-0.075***	0.136***	-0.024***	-0.030***
	(0.002)	(0.002)	(0.001)	(0.0010)
Foreign-Born	0.244***	-0.158***	-0.043***	-0.019***
	(0.001)	(0.001)	(0.001)	(0.0008)
Medicaid Expansion x Foreign-Born	-0.031***	0.007***	0.018***	-0.003***
	(0.002)	(0.002)	(0.002)	(0.001)
State and Year FE	yes	yes	yes	yes
Observations	1,975,594	1,975,594	1,975,594	1,975,594
R^2	0.10174	0.08522	0.00411	0.00827
Within R^2	0.04288	0.02242	0.00165	0.00162

Notes: This play displays result form linear probability models DiD specification 1 for 4 outcome, no control variable included for this result. Heteroskedasticity-robust standard-errors are in parentheses,Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table 6

Effect of Medicaid Expansion on Insurance Coverage by Citizenship Status

	(1)	(2)	(3)	(4)
Effect	Uninsured	Medicaid	Employer Sponsered	Direct Purchase
Treat (Expansion State)	-0.0831***	0.1410***	-0.0230***	-0.0294***
	(0.0018)	(0.0018)	(0.0015)	(0.0011)
FB Citizen × Treat	0.0215***	0.0056	0.0007	-0.0261***
	(0.0036)	(0.0038)	(0.0032)	(0.0024)
FB Non-Citizen × Treat	0.0402***	-0.0620***	0.0214***	-0.0106***
	(0.0034)	(0.0031)	(0.0026)	(0.0017)
Mean of Dep. Var	0.45	0.29	0.22	0.07
Observations	1,975,594	1,975,594	1,975,594	1,975,594
R-squared	0.178	0.195	0.108	0.052
Within R-squared	0.124	0.139	0.106	0.046
Fixed Effects: State, Year	Yes	Yes	Yes	Yes

Note. Table reports coefficients from difference-in-differences models with state and year fixed effects. All models control for age, sex, disability status, poverty status, English proficiency, years in U.S., educational attainment, marital status, race/ethnicity, employment status, state unemployment rate, and immigration policy climate index (IPC). Standard errors are heteroskedasticity-robust and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

FB Citizen and **FB Non-Citizen** interactions capture heterogeneous effects by nativity.

Table 7

Impact of the ACA Expansions on Medicaid: Event Study result

	US-born		Foreign-Born	
Variables	(1)	(2)	(3)	(4)
expansion year = -8	-0.011	-0.017	-0.006	0.003
	(0.029)	(0.034)	(0.009)	(0.010)
expansion year = -7	-0.002	-0.003	-0.011	0.004
	(0.031)	(0.034)	(0.013)	(0.011)
expansion year = -6	-0.002	0.003	-0.010	0.007
	(0.021)	(0.028)	(0.011)	(0.011)
expansion year = -5	-0.024	-0.027	-0.030***	-0.031***
	(0.014)	(0.015)	(0.008)	(0.008)
expansion year = -4	0.013	0.012	0.010	-0.0005
	(0.015)	(0.015)	(0.017)	(0.010)
expansion year = -3	-0.004	-0.003	-0.006	-0.001
	(0.003)	(0.004)	(0.006)	(0.006)
expansion year = -2	-0.003	-0.005	-0.005	0.007
	(0.004)	(0.004)	(0.007)	(0.006)
expansion year = 0	0.082***	0.080***	0.067***	0.064***
	(0.004)	(0.006)	(0.002)	(0.005)
expansion year = 1	0.150***	0.147***	0.144***	0.136***
	(0.011)	(0.012)	(0.013)	(0.011)
expansion year = 2	0.161***	0.159***	0.157***	0.147***
	(0.011)	(0.013)	(0.017)	(0.012)
expansion year = 3	0.167***	0.165***	0.162***	0.153***
	(0.011)	(0.012)	(0.017)	(0.011)
expansion year = 4	0.157***	0.154***	0.182***	0.173***
	(0.010)	(0.011)	(0.018)	(0.016)
expansion year = 5	0.156***	0.157***	0.181***	0.159***
	(0.011)	(0.011)	(0.017)	(0.015)
State FE	yes	yes	yes	yes
Year FE	yes	yes	yes	yes
Controls		yes		yes
Observations	1,585,639	1,585,639	389,955	389,955
R^2	0.06514	0.18775	0.10655	0.18073
Within R^2	0.00628	0.13661	0.00837	0.09070

Note: This table displays Event study coefficients associated with Equation 3. Robust standard errors clustered at the state level reported in parentheses.
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 8

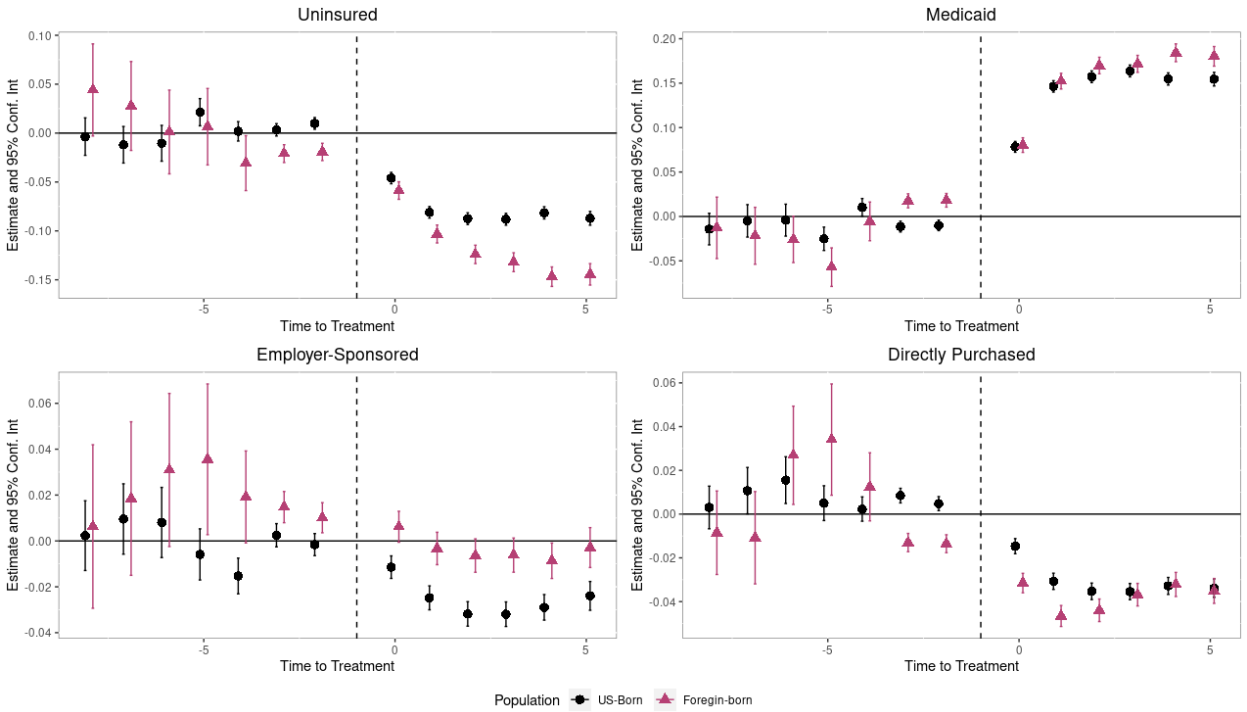
Impact of the ACA Expansions on Uninsured, Event Study result

	US-born		Foreign-Born	
Variables	(1)	(2)	(3)	(4)
expansion year = -8	-0.005	0.005	0.059*	0.057**
	(0.021)	(0.025)	(0.030)	(0.020)
expansion year = -7	-0.014	-0.012	0.043	0.020
	(0.025)	(0.028)	(0.028)	(0.019)
expansion year = -6	-0.012	-0.009	0.009	0.005
	(0.019)	(0.023)	(0.026)	(0.022)
expansion year = -5	0.021*	0.028**	0.007	0.024
	(0.011)	(0.012)	(0.018)	(0.016)
expansion year = -4	-0.001	0.002	-0.012	0.005
	(0.017)	(0.017)	(0.016)	(0.017)
expansion year = -3	-0.001	-0.0003	-0.004	-0.010
	(0.004)	(0.006)	(0.010)	(0.008)
expansion year = -2	0.005	0.005	0.002	-0.016*
	(0.003)	(0.005)	(0.008)	(0.008)
expansion year = 0	-0.052***	-0.051***	-0.035***	-0.023*
	(0.005)	(0.006)	(0.008)	(0.011)
expansion year = 1	-0.090***	-0.086***	-0.069**	-0.059**
	(0.008)	(0.008)	(0.024)	(0.022)
expansion year = 2	-0.100***	-0.098***	-0.076**	-0.058**
	(0.009)	(0.010)	(0.026)	(0.024)
expansion year = 3	-0.099***	-0.097***	-0.090**	-0.073**
	(0.008)	(0.008)	(0.028)	(0.022)
expansion year = 4	-0.094***	-0.092***	-0.099**	-0.082**
	(0.009)	(0.009)	(0.030)	(0.025)
expansion year = 5	-0.100***	-0.100***	-0.099***	-0.065**
	(0.009)	(0.009)	(0.026)	(0.024)
State FE	yes	yes	yes	yes
Year FE	yes	yes	yes	yes
Controls		yes		yes
Observations	1,585,639	1,585,639	389,955	389,955
R^2	0.05930	0.11980	0.08577	0.21568
Within R^2	0.00286	0.06699	0.00172	0.14357

Note: This table displays Event study coefficients associated with Equation 3. Robust standard errors clustered at the state level reported in parentheses.
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Figure 8

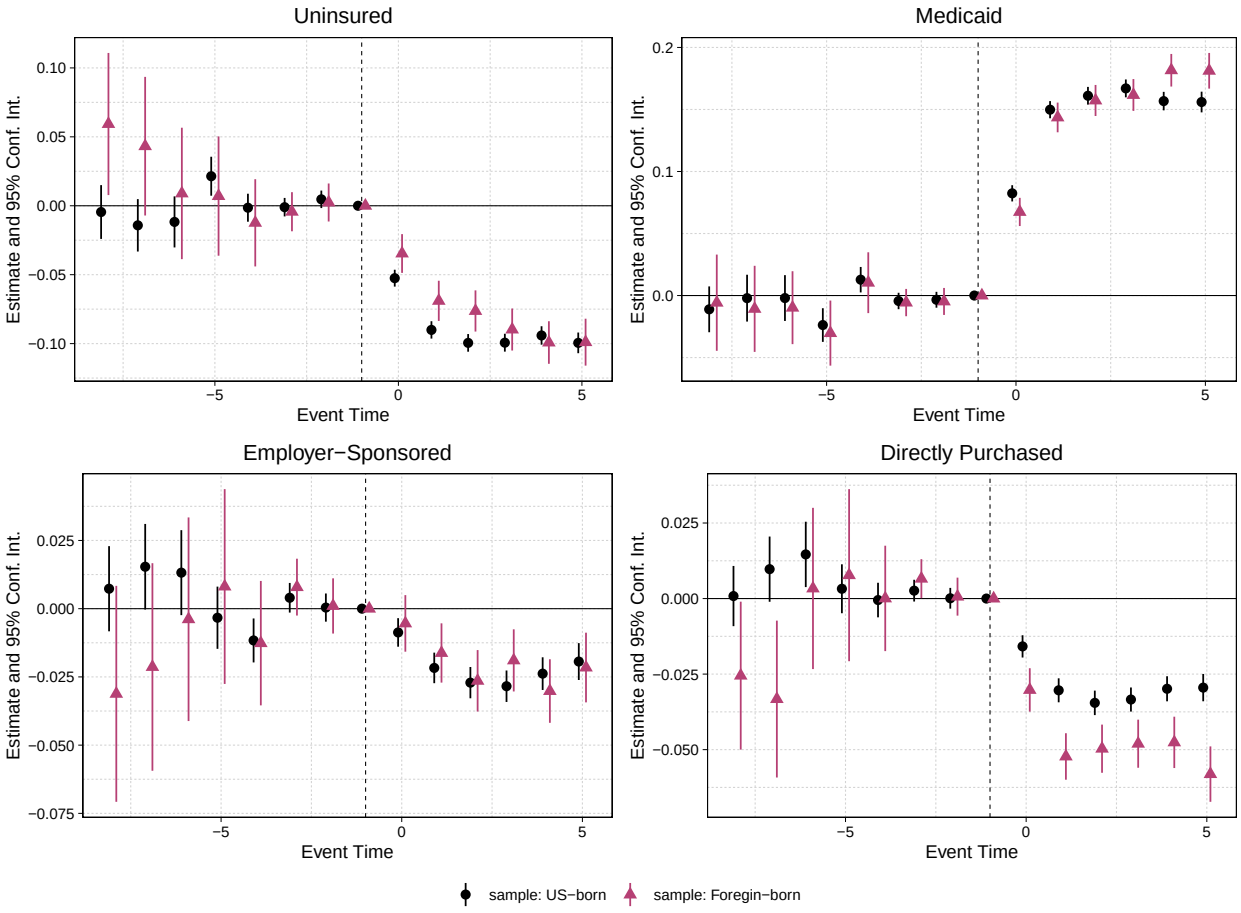
Event-study regression uncontrolled



Note: This figure plots the coefficients from Specification 2 without including any control variables

Figure 9

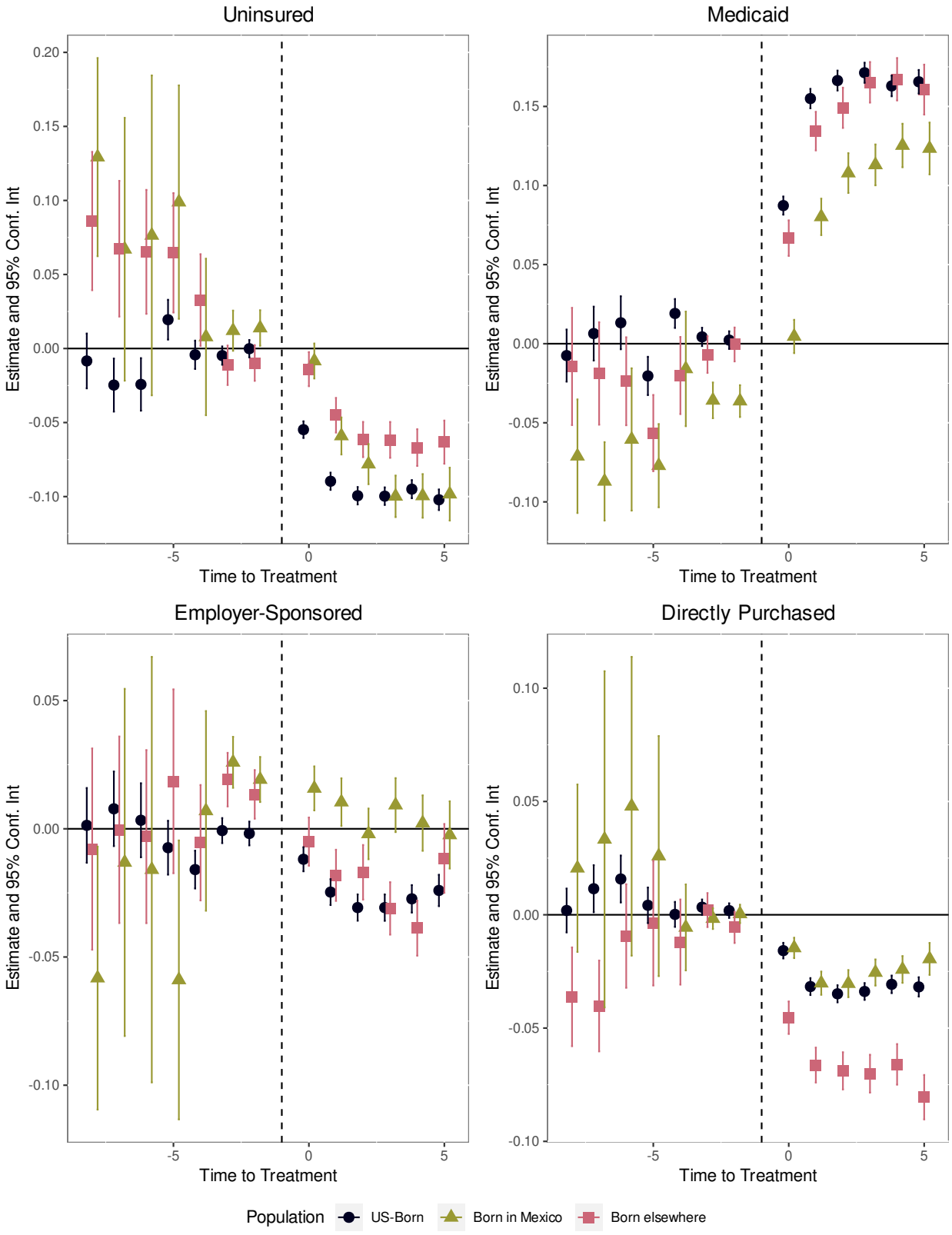
Event-study regression uncontrolled split sample



Note: This figure plots the coefficients from Specification 3, without including any control variables

Figure 10

Event-study regression controlled by Mexican Citizenship



Note: These figures plot event study coefficients that add additional interaction for Mexican Citizenship